



Republic of the Philippines
OFFICE OF THE PRESIDENT
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE



January 28, 2026

MEMORANDUM

TO : THE SENIOR DEPUTY ADMINISTRATOR, DEPUTY ADMINISTRATORS, THE REGIONAL/DEPARTMENT/PROJECT MANAGERS, DIVISION MANAGERS, AND ALL OTHERS CONCERNED

SUBJECT : DISSEMINATION OF NATIONAL IRRIGATION ADMINISTRATION (NIA) RESPONSES TO THE COMMISSION ON AUDIT (COA) ON THE STATUS OF IRRIGATION DEVELOPMENT AND THE NATIONAL IRRIGATION MASTER PLAN (NIMP) 2020-2030

In line with the NIA's continuing engagement with the COA, and to ensure consistency of institutional positions across all offices, copies of the following communications submitted to the COA are hereby disseminated to all concerned offices for reference and guidance:

1. **Letter dated 16 January 2026** transmitting the Agency's *Management Comments on Audit Observation Memorandum (AOM) No. ALL FUNDS-2026-004 (2025)* relative to the review of the Status of Irrigation Development and its alignment with the National Irrigation Master Plan (NIMP) 2020-2030; and
2. **Letter dated 19 January 2026** submitting NIA's response on the *Status of the NIMP 2020-2030 Recommendations* and providing clarification on the *Estimated Total Irrigable Area of 3,128,631 hectares*.

The attached Central Office responses comprehensively address COA's observations and clarify key technical, planning, and policy issues, including but not limited to:

- a. the historical basis and limitations of the long-standing Estimated Total Potential Irrigable Area (PIA);
- b. the distinctions among irrigation area classifications, including Service Area and Firmed-Up Service Area (FUSA);
- c. the purpose, scope, and continuing relevance of the NIMP 2020-2030;
- d. the role of planning, budgeting, and the Public Investment Program (PIP) in irrigation development; and
- e. the transition toward updated, map-based benchmarks through the Regional Irrigation Master Plans (RIMPs).

All concerned offices, **particularly those which have received or may receive similar COA observations**, are hereby **directed to take note of the explanations, positions, and clarifications contained in the attached Central Office responses** and to use the same as reference in the preparation of management replies, audit conferences, and related submissions to COA.

To ensure consistency of reporting, audit defensibility, and alignment with attached communications, all concerned offices are hereby enjoined to closely coordinate with the **Corporate Planning Service via email at cps@nia.gov.ph or by telephone at (02) 8926-5896**.

For information, guidance, and compliance.

ENGR. EDUARDO EDDIE G. GUILLEN
Administrator





Republic of the Philippines
OFFICE OF THE PRESIDENT
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE

January 16, 2026

MR. JOHN GILBERT P. PIO

OIC - Supervising Auditor
Audit Group B - National Irrigation Administration
Commission on Audit

Attention: **MS. FRANIROSE U. MALONG**
OIC - Audit Team Leader
Audit Team 1 - Corporate Fund

SUBJECT: **Management Comments on Audit Observation Memorandum (AOM) No. ALL FUNDS-2026-004 (2025) - Review of the Status of Irrigation Development and its Alignment with the National Irrigation Master Plan (NIMP) 2020-2030**

Sir/Madam:

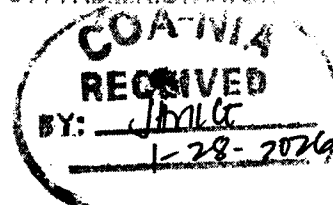
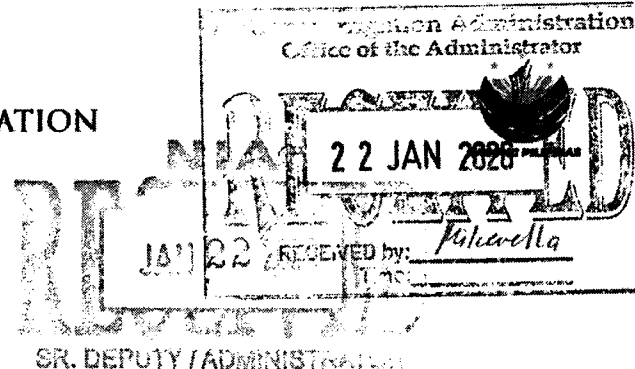
The National Irrigation Administration (NIA) respectfully submits its *Management Comments on Audit Observation Memorandum (AOM) No. ALL FUNDS-2026-004 (2025)* dated 13 January 2026. The Agency appreciates the Commission on Audit (COA)'s constructive observations and recommendations, which are aligned with NIA's ongoing initiatives to strengthen evidence-based planning, data integrity, and institutional accountability.

I. On the Historical Basis of the Estimated Total Irrigable Area (3,128,631 hectares)

The NIA confirms that it has exercised due diligence and exhausted all reasonable means to trace the original internal technical documentation, geospatial datasets, and policy issuances supporting the long-standing Estimated Total Potential Irrigable Area (PIA) of **3,128,631 hectares (ha)**, as reflected in the NIA's *Status of Irrigation Development - Based on Inventory* reports and the NIA website.

Available records indicate that this figure has been consistently cited since at least 2010, and likely earlier; however, no surviving internal reference maps, technical reports, or formal issuances could be retrieved to definitively establish its original derivation. Consultations with long-tenured NIA personnel likewise yielded no verifiable internal documentation. The absence of original reference maps does not imply the absence of a methodology, but rather reflects historical limitations in archival practices and geospatial data management at the time the estimate was produced.

Notwithstanding this limitation, the **technical provenance of the 3.1-million-hectare estimate is substantively documented in the World Bank's *Philippines Irrigated Agriculture Sector Review* (Report No. 9848-PH, 15 April 1992) (Annex A)**. Said report explains that the most frequently cited estimate of potentially irrigable land approximately **3.126 million hectares** originated from NIA's 1990 Corporate Plan and was derived primarily using a **0-3 percent slope criterion**, based on 1:50,000 topographic maps prepared in the 1950s. The World Bank further documented the limitations of this approach, including the absence of systematic consideration of land



conversion, infrastructure constraints, water availability, flood risks, and rapidly changing land use patterns.

Accordingly, while the original reference maps are no longer available, the methodological basis and limitations of the historical estimate are explicitly acknowledged and externally validated, as cited above. This clarification is respectfully submitted to address the observation that the estimate could not be "reliably substantiated," by distinguishing between the absence of archival source maps and the documented existence of the original methodology.

II. On the National Irrigation Master Plan (NIMP) 2020–2030 and the Use of FUSA Projections

The NIA clarifies that the **National Irrigation Master Plan (NIMP) 2020–2030 was intentionally designed to address and correct the limitations of legacy slope-only estimates**, including the 3.1-million-hectare figure. The NIMP adopted updated geospatial basemaps, standardized datasets, expanded slope criteria (0–8 percent), and multi-factor screening that incorporates biophysical, socioeconomic, environmental, legal, and institutional constraints.

With respect to the comparison made in the AOM between the **Estimated Total Irrigable Area** and the **Projected Operational Firmed-Up Service Area (FUSA)** under Scenario 1 of the NIMP, the NIA respectfully submits that these indicators are conceptually and operationally distinct, as defined under *NIA Memorandum Circular No. 146, series of 2022 (Annex B)*, which governs the Agency's official irrigation area classifications:

- **Service Area (Developed Area)** refers to the area of an irrigation system after project completion and turnover for operation, already provided with irrigation and drainage facilities where irrigation and drainage services can be rendered.
- **Converted Area** refers to portions of the service area originally used for agriculture but converted to non-agricultural uses such as commercial, industrial, or residential purposes, normally supported by official conversion documents.
- **Newly Generated Area** refers to areas generated by newly completed projects during the year but not yet ready to receive irrigation services.
- **Permanently Non-Restorable Area** refers to portions of the service area that can no longer be restored or irrigated despite any intervention, such as areas occupied by canals, access roads, or eroded sections.
- **Firmed-Up Service Area (FUSA)** refers to the **net service area** after deducting converted and permanently non-restorable areas from the service area.
- **Operational FUSA** refers to the portion of the firmed-up service area where irrigation facilities are functional and capable of delivering continuous irrigation water.
- **Non-Operational FUSA** refers to areas within the firmed-up service area that are no longer irrigated due to damaged or non-functional facilities and require rehabilitation to restore service.

In this context, the **Projected Operational FUSA under Scenario 1 of the NIMP represents only the operational portion of already developed irrigation systems**, primarily used for palay production and food security modeling. It does **not** represent the country's total potential irrigable area, nor does it equate to physical irrigation development ceilings or expansion targets. Accordingly, direct comparison between Estimated Total Irrigable Area and Projected Operational FUSA figures should be interpreted with due regard to their distinct purposes, definitions, and methodologies.

The NIA further notes that the formulation of the NIMP formally commenced in December 2020 based on the Terms of Reference (TOR) (**Annex C**). Hence, irrigation development data for years prior thereto could not have referenced NIMP-based projections that were not yet available or adopted at the time.

III. On Planning, Budgeting, and the Public Investment Program (PIP)

As explicitly stated in *Section 4.5 (Physical Targets)* of the NIMP 2020–2030, physical irrigation development targets were anchored on the NIA's **Updated Public Investment Program (PIP) 2017–2022** as submitted to the Department of Economy, Planning, and Development (DEPDev, formerly NEDA) during the time of the Plan's formulation. The PIP is a **rolling, annually updated list of priority programs and projects**, serving as the basis for national budgeting, oversight review, and implementation approval by the Department of Budget and Management (DBM) and the Congress (**Annex D**).

In this context, while the NIMP provides a long-term strategic framework, actual irrigation development is *necessarily constrained by annual budget allocations, funding availability, and oversight approvals*. Decisions to accelerate or scale irrigation investments do not rest with NIA alone, but are subject to the collective determination of oversight institutions, including DEPDev and DBM, as well as broader fiscal, climatic, environmental, and social constraints.

The Agency, through the Corporate Planning Service (CorPlan), respectfully clarifies that it is not hesitant to revise the Estimated Total Irrigable Area. However, the NIA consistently recommends that any revision be undertaken only upon **the establishment of a strong, transparent, and technically defensible basis**, in order to prevent a recurrence of issues related to undocumented or insufficiently supported estimates. This approach is aligned with auditability, data integrity, and long-term planning accountability.

It is emphasized that the NIMP 2020–2030 underwent extensive technical validation and peer review prior to its adoption. As required by the Department of Economy, Planning, and Development (formerly National Economic and Development Authority), the NIMP was **peer-reviewed by the World Bank and the University of the Philippines (Annex E)**. The NIA formally responded to all review comments transmitted by DepDev, including recommendations on improving the robustness and adaptability of the Plan.

One critical recommendation from the peer review was the conduct of a midterm updating of the NIMP to account for actual budget allocations, implementation performance, updated datasets, and evolving policy directions. In compliance with this recommendation, the NIA has targeted and planned the conduct of the NIMP Midterm Review within the current planning period, consistent with results-based planning and adaptive management principles.

In addition, the NIMP underwent **rigorous inter-agency review under the Cabinet Assistance System (CAS) of the Office of the President**. During the **57th CAS Meeting held on 1 March 2021**, the NIA presented the *Inputs and Recommendations from various Cabinet Assistance System Agencies on the NIMP 2020–2030*. Comments were received from multiple agencies, including the Office of the Cabinet Secretary, Cabinet Assistance System (CAS), DepDev, DBM, Department of Finance (DOF), Department of Agriculture (DA), Department of Environment and Natural Resources (DENR), Department of Science and Technology (DOST), Department of Trade and Industry (DTI), Department of Energy (DoE), Department of Transportation (DOTr), and the Department of Social Welfare and

Development (DSWD). These inputs were duly considered in the finalization and implementation of the NIMP.

The NIA further clarifies that it **did not claim that the NIMP 2020–2030 is entirely outdated**. Rather, the Agency recognizes that **certain reference datasets used during the formulation of the NIMP have since been superseded** by more recent versions, such as the National Mapping and Resource Information Authority (NAMRIA) Land Cover Map (2020), whereas the NIMP utilized the 2015 Land Cover data that was the most current at the time of formulation. Moreover, development targets cited from the Philippine Development Plan (PDP), which the NIMP referenced, are subject to policy adjustments under succeeding administrations.

As explicitly stated in the NIMP, the inclusion of multiple scenarios and recommendations was intended to ensure flexibility and adaptability to changes in national food security policies and development priorities, particularly considering that the Philippine Development Plan referenced during formulation was valid only until 2022, while the NIMP covers the period up to 2030.

IV. On the Regional Irrigation Master Plans (RIMPs) and Updating of Benchmarks

The preparation of the **Regional Irrigation Master Plans (RIMPs) 2023–2030** operationalizes the NIMP at the regional level and provides a **map-based Net Potential Irrigable Area (Net PIA)**, refined from Gross PIA using updated land cover, environmental, legal, and socioeconomic data. The RIMPs directly address the limitations of legacy estimates by deducting built-up areas, roads, forestlands, protected areas, ancestral domains, fault lines, and other exclusions, consistent with current laws and planning standards.

The ceremonial signing of the **Joint Memorandum Circular (JMC) between the Department of Agriculture and the NIA on 13 January 2026** formally signifies the approval, adoption, and implementation of the RIMPs. Upon consolidation and full approval of RIMP outputs, the updated PIA will **supersede the legacy 3.1-million-hectare estimate** as the official national reference for planning and monitoring, subject to a **managed transition period** to ensure continuity of reporting and avoid disruption of ongoing programs.

V. On Institutional Strengthening and Data Governance

The NIA concurs with COA's recommendations on documentation, institutional memory, and coordination. In line with the NIMP, the Agency has initiated the establishment of a standardized operational GIS environment, centralized digital archiving of planning references, and strengthened inter-agency coordination mechanisms.

These reforms are complemented by a proposed reorganization, recognizing that effective NIMP and RIMP implementation requires dedicated manpower for monitoring and evaluation, geodatabase management, quality assurance, and social and environmental safeguards.

In view of the foregoing, the NIA respectfully submits that:

1. The historical 3,128,631-hectare estimate is traceable to documented legacy methodologies, with limitations clearly identified in authoritative external reviews;

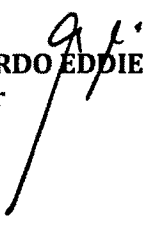
2. The NIMP 2020–2030 and RIMPs 2023–2030 were expressly designed to correct these limitations through updated, map-based, and multi-factor planning; and
3. The Agency is fully aligned with COA's recommendations and is actively implementing reforms that will result in more reliable, transparent, and auditable irrigation development benchmarks.

As discussed above, the adoption of the updated PIA derived from the RIMPs will be implemented with a managed transition period to avoid disruption to ongoing planning, budgeting, reporting, and performance monitoring processes.

In this regard, NIA respectfully requests the Commission on Audit's continued understanding and patience during this transition, as the Agency ensures that the updated benchmarks are fully validated, consistently applied, and properly institutionalized.

Thank you.

Sincerely,


ENGR. EDUARDO EDDIE G. GUILLEN
Administrator

Annex A

Report No. 9848-PH

Philippines Irrigated Agriculture Sector Review

(In Two Volumes) Volume II: Annexes 4 and 5

April 15, 1992

East Asia and Pacific Region

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3. IRRIGABLE LAND AND ITS DEVELOPMENT

A. IRRIGABLE LAND RESOURCES

General

3.1 The primary definition of potentially irrigable land in the Philippines is land on a slope of less than 3%¹. The area defined by slope must then be rectified to take account of soil conditions, surface water and groundwater drainage constraints, the magnitude of flood hazards, and land permanently alienated from irrigation use by road and rail communications infrastructure and urban and rural dwellings and other buildings. Given rapid population growth and its concentration on the plains, increasing amounts of potentially irrigable land will be occupied by settlement.

3.2 Most of the land with slope of less than 3% occurs in valley bottoms of the larger rivers and on the coastal plains. The break in slope from plain to mountains is generally abrupt. Hill streams are in deeply incised, V-shaped valleys with steep gradients and with very limited level land, and there are relatively few major river systems with large associated plain lands. Much of the area is drained from small basins by rivers which flow independently to the sea. Typically, these have steep catchments and proportionately small areas of plain land. The notably extensive areas of plain lands occur in the Cagayan basin of northern Luzon, the Pampanga and Agno basins of Central Luzon, the Bicol plain of southern Luzon, the central plain of Panay Island, the north central plain of Leyte, and the plains of the Agusan, Davao and Cotabato basins in Mindanao.

Estimates of Potentially Irrigable Land

3.3 The most frequently quoted estimate of potentially irrigable land is 3.126 M ha which is given in NIA's 1990 Corporate Plan. This is said to be based on the primary criterion of land with less than 3% slope. The assessment was carried out in the early 1980s based on an analysis of 1:50,000 topographic maps -- with 25 m contours on steep slopes and 5.0 m contours on the plain lands -- dating from the 1950s. These maps indicate areas then predominantly under rice paddies. Examination of some sheets indicates that the definition of the gross irrigable area was sometimes based on the area under paddies (which often extends across the entire plain area to the foothill zone) although in other cases the land slope definition was applied. Adjustments were made within the gross area to account for major settlements and prominent marshes, lakes and hilly zones. However, no account was taken of infrastructure such as roads, nor of the many scattered areas of rural settlement and associated gardens and plantations, areas subject to heavy seasonal flooding, and areas which would have to be dedicated to irrigation, drainage and flood control infrastructure. The definition considered only irrigability of the land, and no account was taken of the availability of water or the costs of delivering it and of providing drainage and flood control facilities. The assessment has not been updated to take account of

¹ This criterion might be violated by including somewhat steeper land if it is already terraced for paddy cultivation and water could be easily delivered.

land taken over for new urban (and rural) settlement since the date of the topographic mapping. Unfortunately, most of the 1:50,000 sheets on which the lands regarded as irrigable were demarcated are now not readily available.

3.4 Another source, ostensibly more detailed than the 1990 Corporate Plan, is the 'Provincial Irrigation Profiles' (PIPs) prepared by NIA in 1989 for the 73 provinces with significant irrigation potential (only the scattered island province of Batanes in the Cagayan Region, Metro Manila, and small islets are omitted). The PIPs are in considerable detail and include estimates of irrigable area, present development in terms of service areas of national and communal irrigation schemes, and new developments categorized as national, communal, SWIM, and pump irrigation projects (Annex 1). However, although the countrywide totals given in the Corporate Plan and as derived by totalling the PIP estimates agree within about 3% (3.126 M ha versus 3.036 M ha), the regional totals have major differences for every region (see Table 3.1). These are so great as to indicate that they must be derived from separate sources.

Table 3.1: Comparison of Estimates of Potentially Irrigable Land from the 1990 Corporate Plan and the 1989 Provincial Profiles

Regions	Potentially irrigable area (ha)		Ratios of Corp. Plan to PIPs
	Corp. Plan	PIPs	
Ilocos	309,810	400,590	0.77
Cagayan Valley	539,710	460,840	1.17
Central Luzon	482,220	441,050	1.09
Southern Tagalog	263,590	238,680	1.10
Bicol	239,650	208,640	1.13
Western Visayas	197,250	171,490	1.15
Central Visayas	50,740	70,270	0.72
Eastern Visayas	84,380	109,450	0.77
Western Mindanao	76,500	70,110	1.09
Northern Mindanao	230,150	249,970	0.92
Southern Mindanao	290,250	241,520	1.20
Central Mindanao	362,080	374,230	0.96
Total	3,126,330	3,036,850	1.03

Source: NIA's Corporate Plan (1990) and PIPs (1989).

3.5 Other estimates of potential irrigable area can be derived from publications of the Bureau of Soils and Water Management (BSWM) of the Department of Agriculture. In 1985, the Bureau initiated the 'Land Resources Evaluation Project' which has produced a detailed report for each province of the land resources and present and potential uses. Perhaps easier to use for regional planning of irrigation will be the results of the Bureau's on-going program to produce a 'Framework for Crop Development and Soil Conservation Planning'. The assessment is based on administrative regions, and each report is supported by regional thematic maps at 1:250,000 scale, of which the 'Rice-Based Cropping System Development and Soil Conservation Planning Guide Map' is particularly useful. The mapping identifies areas which presently have paddy rice in the cropping pattern under seven categories (of which three are irrigated paddy); areas which do not presently include paddy rice in the cropping pattern but are recommended for conversion to paddy; and a class of

present waste lands which are recommended for reclamation for possible cultivation of paddy. As paddy is essentially the only irrigated field crop, this mapping will be very useful for defining the potentially irrigable areas on a sounder basis than slope alone.

Service Areas of Irrigation Projects

3.6 Uncertainties relating to potential irrigable area are also encountered at a detailed project level. Tables A.2 and A.3 summarize estimates of service area from successive implementation stages for a number of World Bank and ADB assisted projects. The results for the nine projects with Project Performance Audit Reports (PPARs) (with multiple loans for the same scheme considered as one project) are summarized in Table 3.2 with service area estimates also compared to actual 1989 wet season (WS) irrigated area.

Table 3.2. IBRD- and ADB-Supported Projects with PPARs: Service Area Estimates

	Service Area (ha)			WS Area	
	Appraisal	PCR	PPAR	Actual 1989	1989 (ha)
IBRD Projects (5)	283,170	274,601	241,331	234,001	202,944
ADB Projects (4)	39,420	31,590	31,387	27,920	23,608
Total (9)	<u>322,590</u>	<u>306,191</u>	<u>272,718</u>	<u>261,921</u>	<u>226,552</u>
% of Appraisal Estimate	100	95	85	81	70

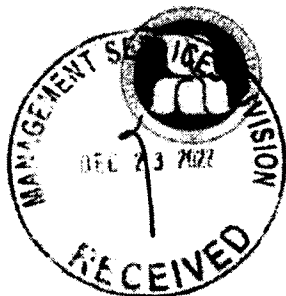
Source: Attachment 1, Tables A.2 and A.3.

3.7 Service area figures are adjusted periodically and the differences indicated by Table 3.2 are to some extent arbitrary (that between the PCR and PPAR totals for instance was almost entirely due to adjustments made by the PPAR for MARIIS). Nevertheless, a 20% difference between actual service area and design estimates has been generally confirmed by a recent NIA review of six existing systems which involved detailed field surveys and hydrological reassessments. Furthermore, the difference between revised service area and actual wet season irrigated areas suggests if anything that 20% may be too low. NIA is currently completing a systematic 'firming' up of all service areas in national systems based on detailed cadastral mapping financed under Bank-supported projects and this should provide more definitive estimates.

3.8 Similar reductions in service area, and wet season irrigated areas significantly below expectations, have also been encountered in respect of communal irrigation schemes. As a result, improved procedures for estimating service area have been included under the recently-approved Bank-supported second communal project (CIDP II).

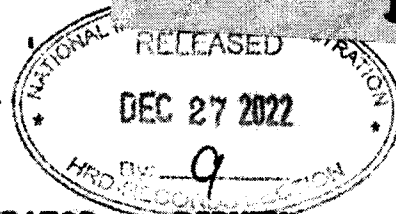
Reassessment of Irrigation Potential

3.9 Significant reductions in wet season irrigated areas for both NISs and CIS can have a potentially major impact on economic returns (Chapter 4). The 'firming up' of NIS areas and revised procedures for CIS indicate that NIA recognizes this problem. To complement these project-level activities, an



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NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE

MEMORANDUM CIRCULAR No. 146
Series of 2022



Annex B

RECEIVED

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TO : THE SENIOR DEPUTY ADMINISTRATOR, DEPUTY ADMINISTRATORS, THE REGIONAL/DEPARTMENT/PROJECT MANAGERS, DIVISION MANAGERS, AND ALL OTHERS CONCERNED

SUBJECT : GUIDELINES FOR THE COLLECTION, UPDATING, AND VALIDATION OF IRRIGATION DATA IN THE ONLINE INVENTORY OF IRRIGATION SYSTEMS (OIS)

I. RATIONALE

Based on the study conducted by the Philippine Institute for Development Studies (PIDS), entitled *Revitalizing Philippine Irrigation: A Systems and Governance Assessment for the 21st Century*¹, there is a need to improve the irrigation data of National Irrigation Systems (NIS) and Communal Irrigation Systems (CIS) under NIA, Other Government Agencies (OGA) assisted by Department of Agriculture (DA) and Private Irrigation Systems (PIS) by utilizing geo-referenced data for the estimation of potential irrigable areas to account for the irrigation development and the expansion of residential, commercial, and industrial uses of land. The study also recommended that NIA should increase its collaboration with DA and LGUs in identifying potential irrigation projects relevant to the provinces, particularly in terms of commodities and locations to support.

The abovesaid findings are also aligned with the recommendations of the National Irrigation Master Plan (NIMP) 2020-2030 to improve NIA's current database using new spatial or map-based data that will prioritize mapping not just in operations and maintenance but also in planning. Area declaration in the reports shall be based on the approximate computed area from boundary delineation through maps. Area in the inventory should always be validated in the field.

The Master Plan also recommended the maintenance enhancement and operation plan for irrigation systems including asset management. These involved reviewing existing O&M management methods, practices, and monitoring systems, and proposing methods and strategic plans for improved O&M systems. Hence, reviewing the monitoring mechanisms of the Agency is necessary to update current datasets and include other sources such as from other relative agencies and geospatial maps.

Following this, a coordination meeting was conducted last September 20, 2022, among NIA, the Bureau of Soils and Water Management (BSWM), and the Bureau of Agricultural and Fisheries Engineering (BAFE) regarding the validation of irrigation development report for irrigation systems under OGA. The DA-BSWM shall reconcile with the existing NIA Inventory System the data under OGA for Small Scale Irrigation Projects implemented by BSWM and DA-Regional Field Offices based on the Status of Irrigation Development ending December 31, 2021. While NIA shall coordinate with other government offices implementing irrigation projects not covered under the DA Banner Programs.

¹ Briones, R. M. (2021). *Revitalizing Philippine Irrigation: A Systems and Governance Assessment for the 21st Century*. Quezon City: Philippine Institute for Development Studies.



In view of the above-said recommendations, guidelines on the collection, updating, and validation of irrigation data in the Online Inventory of Irrigation Systems (OIIS) are hereby established.

II. COLLECTION FROM SOURCES

The Online Inventory of Irrigation Systems was developed to facilitate ease of submission, consolidation, and verification of the inventory data from the various sources for data/report generation. For guidance and reference, all concerned offices are instructed to adopt the OIIS User's Manual as updated on the online system (<https://inventory.nia.gov.ph>).

The reporting template/form in the Online Inventory is hereby revised based on the comments and recommendations of BSWM and BAFE, NIA Regional Offices, and requirements of the NIMP 2020-2030 (Please see "Annex A"). This shall ensure all data requirements needed will be captured in the monitoring of the Agency's performance and the overall irrigation development in the country. Improved data collection and management are required, given that data adequacy and quality were found to be the constraints to the proper estimation of irrigable areas.

To streamline data sources, continuous coordination and collaboration with the field offices, Local Government Units (LGUs), and concerned DA bureaus and attached corporations are vital to effectively monitor and update the inventory data. The following matrix below shall serve as the basis for the data collection:

Table 1. Data Collection Matrix

Required Data	Source Agency
National Irrigation Systems	NIA
Communal Irrigation Systems	- NIA - CARP-IC
Private Irrigation Systems	- NIA - LGUs through Agricultural Offices
Other Government Agency (OGA) Assisted Irrigation Systems / DA-Assisted Irrigation Systems	- DA-BSWM (including DA-Regional Field Office) - National Tobacco Administration (NTA) - Philippine Coconut Authority (PCA) - Sugar Regulatory Administration (SRA)

III. UPDATING OF DATA AND TIMELINES

The updating of the online inventory of irrigation systems shall be conducted annually. Said online updating shall be done in coordination with the System Management Division (SMD), Management Information Division (MID), Regional Office (RIOs), Project Management Offices (PMOs), Irrigation Management Offices (IMOs), DA-BSWM, NTA, PCA, and SRA.

As adopted from Memorandum Circular No. 47, s. 2012, in updating the inventory of the area of an irrigation system at the end of any particular year under review, the actual

area accomplishment for that particular year (Current Year Program and Previous Year's Carry-Over Accomplishments) shall be incorporated into the said inventory:

- a) **Service Area** – The area of an irrigation system after the project completion and turnover for the operation that is already provided with irrigation and drainage facilities where irrigation and drainage service could already be rendered.
- b) **Converted Area** – That part of the service area of an existing which was originally used for agriculture and is converted to non-agricultural use like: commercial, industrial, subdivision, etc. These are usually supported by documents for conversion.
- c) **Newly Generated Area** – areas of newly completed projects that generated areas this year but are not yet ready to receive irrigation
- d) **Permanently Non-Restorable Area** – That part of the service area of an existing system which can no longer be restored/irrigated even after any intervention that NIA may apply, e.g. areas covered by canals, access/service roads, the portion of service areas that were eroded, etc.
- e) **Firmed-Up Service Area (FUSA)** – The net service area of an irrigation system where converted areas and permanently non-restorable areas were deducted from the service area.
- f) **Operational FUSA** – refers to the area within the firmed-up service area of an irrigation system where the constructed facilities are functional or operational to allow a continuous flow of irrigation water to serve the area.
- g) **Non-Operational FUSA** – refers to the affected area within the firmed-up service area of an irrigation system not anymore irrigated because the constructed facilities thereat are either damaged, destroyed or are non-functional that irrigation water cannot flow to the said area, hence the need to repair the damaged structure(s) to restore its functionality and capacity to irrigate.
- h) **Restored Area** – That part of the FUSA or the non-operational area of an irrigation system where irrigation and drainage service could be resumed after the repair of damaged facilities.
- i) **Irrigated Area** – This refers to the area within the operational area of an irrigation system served during the respective cropping seasons: wet season, dry season, and 3rd crop.
- j) **Harvested Area** – The actual portion of the irrigated area of an irrigation system planted to crops during the respective cropping seasons: wet season, dry season and 3rd crop, and where harvest was made.
- k) **Benefited Area** – The actual portion of the irrigated area of an irrigation system planted to crops during the respective cropping seasons: wet season, dry season and 3rd crop and where reasonable harvest (40 cavans per hectare and above) was made.

In addition to the above data, all field names in the inventory form in the excel file shall have to be updated:

- **Year Covered** - indicate the particular year under review

- **Category** - indicate the code number, as follows: for National Irrigation System, for Communal Irrigation System, for Private Irrigation System, and for Other Government Agency (OGA) - Assisted
- **Region** - indicate the code number for the Region
- **Province** - indicate the code number for the Province
- **Municipality** - indicate the name of the municipality covered by the System. Spelling for the name of the municipalities should be consistent with the most recent published in the Philippine Standard Geographic Code (PSGC). No abbreviation is allowed.
- **District** - indicate the code number for the Congressional District
- **Rice Area per year** - indicate the area planted for rice, the average yield for the cropping seasons (wet, dry, and ratooning) including third cropping, if any
- **Sugarcane/Coconut/Banana/Corn/Pineapple/Mango/Tobacco per year** - indicate the area planted per crop and average yield per cropping season
- **Fishpond** - indicate the surface area covered
- **Canals** - indicate the length in kilometers reported for lined and earth canals including HDPE pipes categorized by main and lateral canals
- **Service/Access Roads** - indicate the length in kilometers reported for each type of service/access road categorized by concrete, gravel, and earth
- **Type of Headworks (Dam)/Irrigation Scheme** - indicate the code number, as follows: 1 - Reservoir; 2 - Gravity; and 3 - Pump. *Specify a specific type of diversion for gravity and pump.*
- **Name of dam** - indicate the name of the dam the irrigation system is deriving water from
- **Water Permit** - indicate the volume of water approved in LPS (liters per second), date granted, and annual dues.
- **Source of Water Supply (River)** - indicate the name of the river, creek, or other source/(s) of water supply
- **Point of Diversion** - indicate the grid coordinates of the Latitude and Longitude in decimal degree format.
- **Total No. of Farmer-Beneficiaries** - indicate the total number of farmer-beneficiaries of the system by municipality
- **Total No. of Irrigators' Associations (IAs)/Farmers' Association (FA)** - indicate the total number of IAs/FAs of the system by municipality
- **Remarks** - indicate data source, reasons for the non-operability of the system, for the decrease in areas, the status of ongoing projects, etc.
- **Resp_Region** - indicate the code number of the region responsible for the operation and maintenance of the system

- **Resp_IMO** - indicate the code number of the IMO responsible for the operation and maintenance of the system

The SMD shall spearhead the updating and validation of data entry of said inventory. The MID-CorPlan shall serve as the data bank/clearing house for this inventory and undertake the computer programming of the required outputs.

Other required documents such as the Turnover Documents and the Farmland Geographic Information System (GIS) generated maps are compiled in one (1) Portable Document Format (PDF) file per system which shall be uploaded to the online inventory. For ongoing projects, the Monthly Progress Report (MPR) is required to be included.

IV. RECONCILIATION AND VALIDATION

For ending December 31, 2022, the review and reconciliation of the submitted updated inventory data shall be done in coordination with the BSWM, BAFE SRA, NTA, PCA, and NIA Central Office spearheaded by the Operations Department and Corporate Planning Services with the participation of the Regional Offices including UPRIIS and MARIIS with their respective IMOs/Divisions and PMOs within their coverage area to capture all irrigation data and establish coordination with all irrigation implementing agencies.

For the succeeding years, the NIA Regional Offices are directed to coordinate and collect the updated data from concerned DA Offices at the field level every cropping season.

In view of this, the NIA Regional Offices including UPRIIS, and MARIIS are directed to conduct their reconciliation workshop with their respective IMOs/Divisions and PMOs within their coverage area prior to the overall review in the NIA Central Office and to conduct validation meetings for the source of data with LGUs and DA-Regional Agricultural Engineering Divisions (RAEDs) and Field Operations Division for common reference.

The online inventory shall be opened for the whole year for the encoding of data. Data reconciliation shall start at the Regional level in December of the particular year. By January of the succeeding year or last cropping, validation shall be conducted headed by NIA Central Office participated by Regional Offices and concerned offices under DA.

V. EFFECTIVITY

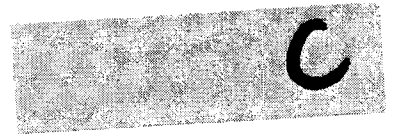
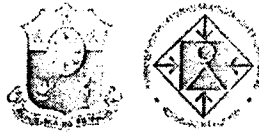
Any provisions that are inconsistent with this issuance shall be deemed superseded. This memorandum shall take effect immediately.

For compliance.


ENGR. EDUARDO EDDIE G. GUILLEN
 Acting Administrator

Date 23 DEC 2022

Annex C



REPUBLIC OF THE PHILIPPINES

NATIONAL ECONOMIC AND DEVELOPMENT AUTHORITY

11 December 2020

CERTIFICATE OF COMPLETION AND ACCEPTANCE/SATISFACTORY SERVICE

This is to certify that the **University of the Philippines Los Baños Foundation Inc. (UPLBFI)**, the Consulting Firm commissioned by the National Economic and Development Authority (NEDA) for the Consulting Services for the **Formulation of the National Irrigation Master Plan (NIMP)**, has satisfactorily completed all its obligations under the Contract Agreement, with a contract amount of **Forty Million Seven Hundred Forty-Two Thousand One Hundred Thirty-Six Pesos (PH 40,742,136.00)**, in accordance with the Terms of Reference (TOR).

This Certification is being issued for whatever legal purpose this may serve.


KATHLEEN P. MANGUNE
Director, Infrastructure Staff

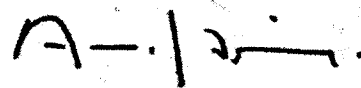
Annex D



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF ECONOMY, PLANNING, AND DEVELOPMENT

MEMORANDUM

FOR : ALL NATIONAL GOVERNMENT AGENCIES, GOVERNMENT-OWNED OR -CONTROLLED CORPORATIONS, GOVERNMENT FINANCIAL INSTITUTIONS, OTHER NATIONAL GOVERNMENT OFFICES AND INSTRUMENTALITIES, AND STATE UNIVERSITIES AND COLLEGES

FROM : ARSENIO M. BALISACAN, PhD  Secretary 

SUBJECT : Updating of the Public Investment Program (PIP) 2023-2028 and Formulation of the Three-Year Rolling Infrastructure Program (TRIP) for Fiscal Years (FYs) 2027-2029 as Input to FY 2027 Budget Preparation: Authorization of Agency PIP/TRIP Focal(s)

DATE : October 8, 2025

1. This is to request your offices to update your existing PIP/TRIP Focal user accounts or nominate your new PIP/TRIP Focal(s), who will be authorized to access the PIP Online (PIPOL) System and will encode the data requirements for priority programs and projects (PAPs) to be included in the updating of the PIP 2023-2028 and formulation of the TRIP for FYs 2027-2029 as input to FY 2027 budget preparation.
2. For this year's PIP updating, to further strengthen the planning-programming-budgeting process, all agencies and offices shall be required to provide information on the agency PAPs included in the PIP and their corresponding (i) annual budget proposal for FY 2026 submitted to the Department of Budget and Management (DBM), and (ii) budgetary allocation in the National Expenditure Program (NEP) for FY 2026 and Government Appropriations Act (GAA) for FY 2025, for submission to the concerned committees of the Economy and Development Council, the Senate, and the House of Representatives in relation to the national budget review and approval process. Please be guided by the **Indicative Calendar of Activities (Attachment 1)** for the milestones of the PIP updating and TRIP formulation exercise.
3. In submitting priority PAPs and Agency Regional-National Investment Projects (ARNIPs)¹, agencies shall use the PIPOL System, which is accessible through <https://pipol.depdev.gov.ph>. The PIPOL System is a web-based project database system which manages data entries and updates on PAPs, and includes a report generation function. Access to the system will only be granted to authorized PIP/TRIP Focals of agencies/offices.
4. To ensure the accountability, accuracy, and timeliness of the submission of agency inputs to DEPDev, we request the agencies to (i) **update their existing agency**

¹ For the RNIP agencies, as follows: Department of Agriculture, Office of the President-National Irrigation Administration, Department of Education, Department of Environment and Natural Resources, Department of Information and Communications Technology, Department of the Interior and Local Government, Department of Health, Department of Science and Technology, Department of Transportation, Department of Social Welfare and Development, Department of Energy, Department of Human Settlements and Urban Development, and Department of Public Works and Highways.



Updating of the Public Investment Program (PIP) 2023-2028 and Formulation of the Three-Year Rolling Infrastructure Program (TRIP) for Fiscal Years (FYs) 2027-2029 as Input to the FY 2027 Budget Preparation:
Authorization of Agency PIP/TRIP Focal(s)

- PIP/TRIP Focal User Account(s) or (ii) nominate new PIP/TRIP Focal(s), as applicable.
- In authorizing PIP/TRIP Focals, the agency should endorse ***only one (1) Lead Focal with at least a Director IV level position*** and up to ***four (4) technical staffs including at least one Budget Officer*** authorized to access the PIPOL System and encode the data requirements and/or provide updates for PAPs to be included in the PIP/TRIP. Please be guided by the **Terms of Reference (TOR) (Attachment 2)** for the duties and responsibilities of authorized agency PIP/TRIP Focal(s).
 - Should you wish to (i) update your existing PIP/TRIP Focal User Account(s), please use the **Agency User Management Module of the PIPOL System**. On the other hand, for agencies with no existing PIP/TRIP Focal(s), kindly use the **Sign-up Page of the PIPOL System** to nominate your focal(s). Attached, for reference in the authorization of agency PIP/TRIP Focal(s), is the **Procedures for the Authorization of Agency PIP/TRIP Focal(s): (A) Updating of Existing PIP/TRIP Focal User Account(s), and (B) Nomination of New PIP/TRIP Focal(s) (Attachment 3)**.
 - All requests for the authorization of PIP/TRIP Focal(s) shall **only** be submitted through the PIPOL System (Sign-up Page and User Management Module) **until October 24, 2025**.
 - By authorizing your PIP/TRIP Focal(s), it is understood that all information submitted in the PIPOL System regarding PAPs on or before the specified deadline are **complete, correct, compliant, and duly endorsed/approved by both the Head of Agency and, if applicable, the Head of Parent Agency**.
 - For State Universities and Colleges (SUCs), please coordinate with the Commission on Higher Education through the Office of Planning, Research, and Knowledge Management for the authorization of SUC PIP/TRIP Focal(s).
 - For queries and technical concerns, you may contact the PIP and TRIP Secretariats through the following email addresses:

PIP Secretariat	&PIP@depdev.gov.ph
TRIP Secretariat	tripsecretariat@depdev.gov.ph

Thank you.

Attachments:

- Indicative Calendar of Activities for the Updating of the PIP 2023-2028 and Formulation of the TRIP for FYs 2027-2029 as Input to FY 2027 Budget Preparation
- TOR of the Authorized PIP/TRIP Focal(s)
- Procedures for the Authorization of Agency PIP/TRIP Focal(s): (A) Updating of Existing PIP/TRIP Focal User Account(s), and (B) Nomination of New Agency PIP/TRIP Focal(s)
 - Annex A: PIPOL System-Generated Authorization Form for Agency PIP/TRIP Focal(s)
 - Annex B: DEPDev Letter to Head of Agency and Head of Parent Agency on the Authorization of Agency PIP/TRIP Focal(s)





REPUBLIC OF THE PHILIPPINES
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE

Annex E Annex D

E

April 21, 2021

MR. JONATHAN L. UY
OIC-Undersecretary
Investment Programming Group
National Economic and Development Authority (NEDA)
12 St. J. Escriva Drive, Ortigas Center, Pasig City

**Subject: Updates on the National Irrigation Master Plan (NIMP) 2020-2030
for the NEDA Board Committee on Infrastructure - Technical Board
(INFRACOM-TB)**

Dear Undersecretary Uy:

In reference to your memorandum dated April 12, 2021, on the agreements reached by the INFRACOM-TB during its virtual meeting held on April 6, 2021, we are respectfully submitting NIA's updates and actions taken on the NIMP during the Cabinet Assistance System (CAS) meetings.

In the 57th CAS Meeting conducted last February 10, 2021, NIA presented the NIMP and requested the body's support in its implementation, as attached in Annex A. Relative to this, Cabinet Secretary Karlo Alexei B. Nograles emphasized the importance of convergence from different agencies and sought their support in having the NIMP approved by the Cabinet and President. Inputs and comments from CAS Agency Representatives were requested by the Cabinet Secretary for NIA's presentation to their succeeding meeting.

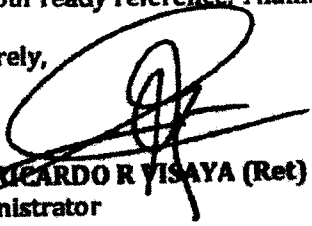
Hence, at the 58th CAS Meeting last March 1, 2021, NIA again presented the additional inputs and recommendations from respective agencies, as attached in Annex B. The Department of Agriculture (DA) Undersecretary Leocadio S. Sebastian stated his support especially on the investment plan that considers many future technologies and some of his suggestions for the implementation of the NIMP which were also discussed in the presentation.

Moreover, during the formulation of the NIMP, the DA Bureau of Agricultural and Fisheries Engineering (BAFE) and Bureau of Soils and Water Management (BSWM) participated in the activities and submitted their comments on the draft NIMP afterwards submitted to NEDA, as attached in Annex C and were considered by the consultants in the finalization of the Master Plan.

Futhermore, concerning the Department of Budget and Management (DBM) and Department of Finance (DOF) issues and concerns on the NIMP materials particularly on the estimated cost of few projects and the target budgetary requirement yearly of the plan, respectively, NIA conducted a virtual discussion with both agencies on the reasons and references of those data formulated with the consultants on the NIMP. The representatives from DBM and DOF expressed their support, but their earlier recommendations still stand. NIA's separate responses were submitted to DBM and DOF with consideration of NEDA's Inputs to proceed with the Agency's further action, as shown in Annex D.

For your ready reference, Thank you.

Sincerely,


GEN. RICARDO R. VISAYA (Ret)
Administrator



REPUBLIC OF THE PHILIPPINES
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE

June 14, 2021

MR. RODERICK M. PLANTA
Assistant Secretary
Investment Programming Group
National Economic and Development Authority (NEDA)
12 St. J. Escriva Drive, Ortigas Center, Pasig City

Subject: Response on Matters Requested for Updates on National Irrigation Master Plan (NIMP) 2020-2030 submitted to NEDA Board Committee on Infrastructure-Technical Board (INFRACOM-TB) and the Peer Review with Agency Inputs

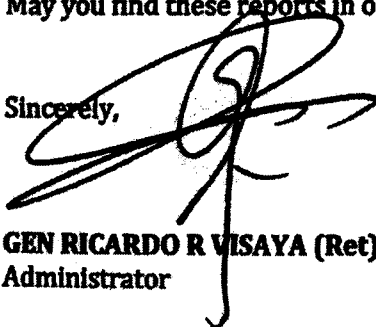
Dear Assistant Secretary Planta:

In reference to your letter dated June 10, 2021, we are hereby submitting copy of the compliance to the instructions of the INFRACOM-TB in a letter dated April 21, 2021, addressed to NEDA Undersecretary Jonathan L. Uy with the NIA's response/action taken to address the issues/concerns earlier raised in the discussion of the master plan at the Cabinet Assistance System (CAS). We already considered and harmonized comments from CAS Agency representatives including inputs from the Department of Agriculture (DA) which were reported at the 58th CAS Meeting last March 1, 2021, see attached **Annex A**.

Relative to the Peer review comments from Ms. Rong Qian of World Bank and Dr. Gilbert Llanto from the University of the Philippines, we are also submitting the agency responses/inputs, see attached **Annex B**.

May you find these reports in order and for your perusal and reference.

Sincerely,


GEN RICARDO R. VISAYA (Ret)
Administrator



REPUBLIC OF THE PHILIPPINES

NATIONAL ECONOMIC AND DEVELOPMENT AUTHORITY

04 August 2021

DR. GILBERTO M. LLANTO

University of the Philippines (UP) -

Public Administration Research and Extension Services Foundation

2F NCPAG Library, R.P. De Guzman St.

UP Dilliman, Quezon City

SUBJECT: Peer Review of the National Irrigation Master Plan (NIMP)

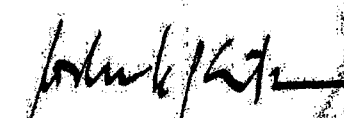
Dear Dr. Llanto:

This pertains to your comments on the NIMP. Your insights are very much appreciated as these are invaluable in enhancing the master plan towards its implementation.

Please be informed that the peer review comments were shared to the National Irrigation Administration (NIA) as the lead agency and the University of the Philippines Los Baños Foundation, Inc. (UPLBFI) as the Consultant of the subject Master Plan. Attached as Annex are the consolidated responses of UPLBFI, NIA, and NEDA to your comments.

Thank you for your invaluable effort in peer-reviewing the Master Plan. For queries or clarifications, your staff may contact the Water Resources Division (WRD) of the Infrastructure Staff (IS) through electronic mail at IS-WRD@neda.gov.ph.

Very truly yours,

 Digitally signed by
Planta Roderick
MARI

Assistant Secretary for Infrastructure Development

Peer Review of the National Irrigation Master Plan (NIMP) Peer Review of the National
Irrigation Master Plan (NIMP)

Page 1 of 1

ANNEX. Consolidated UPLBFI-NIA-NEDA Response to Peer Review Comments on the National Irrigation Master Plan (NIMP)

Comment	UPLBFI-NIA-NEDA Response/Remarks
NIA, the reduction in its work force and inability to hire good staff (many hold casual positions). The section on NIA restructuring should highlight the negative impact on the institution of appointing incompetent administrators. It needs to present a clear organizational structure for NIA and provide the contours of how NIA could be revamped and rehabilitated.	accordance with the prescribed guidelines set forth by the Governance Commission for GOCCs (GCG), to be approved first by the NIA Board prior to submission to GCG for approval.
6. If there is scope for a rewriting of the NIMP perhaps the following should be discussed. NIMP is supported by four pillars: • Clearly articulated policy (rice self-sufficiency? Crop diversification with stress on HVC? Two track policy of rice and HVC?); • An effective institution to orchestrate Plan implementation with capacity to plan and program investments, free of political interference; with resident management and technical expertise; high degree of professionalism; • Feasible investment plan, and not a mere laundry list of irrigation projects all over 57 provinces (high yielding and low yielding; competitive and uncompetitive lumped together for political reasons, perhaps); • Resources to fund the investment projects; budgetary allocation, ODA funding and private participation (with note that it is hard to entice PSP with a free irrigation service act constraining willingness to invest); and • Investment in good data to guide policy, planning and investment programming.	The updating/review of the NIMP may be conducted during the mid-term of the plan (i.e., in 2025), as recommended by the World Bank expert, or earlier once NIA has already established its new organizational structure and the devolved functions are assumed by the LGUs by 2024. In view of the devolution of communal irrigation from NIA to LGUs, the latter shall be involved in the updating preparations.



REPUBLIC OF THE PHILIPPINES

NATIONAL ECONOMIC AND DEVELOPMENT AUTHORITY

04 May 2021

GEN. RICARDO R. VISAYA (Ret.)

Administrator

National Irrigation Administration (NIA)

NIA Bldg., National Government Center

Epifanio de los Santos Avenue, Diliman, Quezon City

SUBJECT: Formulation of the National Irrigation Master Plan (NIMP)

Dear Administrator Visaya:

May we request NIA's appropriate action and response on the following agreements reached by the NEDA Board Committee on Infrastructure – Technical Board (INFRACOM-TB) during its 6 April 2021 meeting wherein the subject formulation of the NIMP was discussed.

As initially coordinated by the NEDA Infrastructure Staff (IS) with your Office, the INFRACOM-TB requested NIA to:

- Report on the response/action taken to address the issues/concerns earlier raised in the discussion of the NIMP at the Cabinet Assistance System (CAS),
- With the NIMP also extensively discussed at the CAS, harmonize efforts undertaken thus far at both the CAS and the INFRACOM-TB, in order to streamline the process moving forward with elevating the NIMP at the Cabinet Level, and,
- Ensure complementarity and alignment of efforts of the NIMP with that of the overall agricultural program of DA.

Formulation of the National Irrigation Master Plan (NIMP) Formulation of the National Irrigation Master Plan (NIMP)

Page 1 of 2

NEDA also requested assistance from local and international experts namely Dr. Gilbert Llanto from the University of the Philippines and Mr. Rong Qian from the World Bank in conducting a peer review on the Master Plan. Attached, for your perusal, is a copy of the results of said peer review.

We look forward to receiving your response to the abovementioned INFRACOM-TB instructions and peer review comments. Also, with a view of integrating our collective response, may we likewise invite you and/or your staff to a meeting on **07 May 2021 at 10 AM** to discuss these.

For queries or clarifications, your staff may contact the Water Resources Division (WRD) of NEDA-IS through electronic mail at WRD@neda.gov.ph.

Thank you.

Very truly yours,

 Digitally signed
by Planta
RODERICK M. PLANTA Roderick Mari
Assistant Secretary for Infrastructure Development

Attachments: (1) Comments of World Bank on the NIMP
(2) Comments of Dr. Gilbert Llanto of UP on the NIMP



REPUBLIC OF THE PHILIPPINES
COMMISSION ON AUDIT
Corporate Government Audit Sector
Cluster 5 – Agricultural and Natural Resources

Office of the Auditor
Audit Group B – National Irrigation Administration

AOM No. : ALL FUNDS-2026-004 (2025)
Date : January 13, 2026

AUDIT OBSERVATION MEMORANDUM

FOR : **Engr. EDUARDO EDDIE G. GUILLEN**
Administrator
National Irrigation Administration (NIA)
EDSA, Quezon City

ATTENTION : **Engr. ROBERT C. SUGUITAN**
Senior Deputy Administrator
Office of the Senior Deputy Administrator

Engr. REYNE B. UGAY
Officer-in-Charge
Office of the Deputy Administrator for Engineering and Operations
Sector

Engr. FERDINAND G. SALES
Acting Department Manager
Office of the Corporate Planning Services

PATRICK R. RILLON
Acting Division Manager
Planning and Programming Division

SUBJECT : **Review of the Status of Irrigation Development – Based on Inventory Reports and its alignment with the National Irrigation Master Plan (NIMP) 2020- 2030**

The NIA's reported Estimated Total Irrigable Areas of 3,128,631 hectares, as reflected in the Status of Irrigation Development – Based on Inventory of NIA Report, cannot be considered reliable, as Management was unable to provide the basis or technical references used in deriving the estimate, which undermines the credibility of irrigation development data used for performance evaluation and public reporting of the NIA. Further, the estimate is inconsistent with the targets and assumptions set forth in the National Irrigation Master Plan (NIMP) 2020 -2030.



1. In line with its mandate under Republic Act (RA) No. 3601, as amended by Presidential Decree Nos. 552 and 1702, the National Irrigation Administration (NIA) is tasked with supporting the country's goals of rice self-sufficiency and poverty reduction through irrigation development. This includes the sustainable development and management of water resources for irrigation, aligned with the agricultural development program of the Government.
2. The NIA publishes the *Status of Irrigation Development – Based on Inventory*, which provides the approved data on the development of irrigation systems in the Philippines. These reports are publicly accessible on the NIA's official website and cover each calendar year from 2011 to 2024.
3. Based on the report, one of the key elements typically contained in this report is the **Estimated Total Irrigable Area** (in hectares) for the country and by region, the estimated total irrigable area for CYs 2011 to 2024 are presented in Table 1.

Table 1 - Estimated Total Irrigable Areas from CYs 2011 to 2024

Year Covered	Estimated Irrigable Areas
2011-2012	3,126,340
2013-2015	3,109,609
2016-2025	3,128,631

4. In view of the farmers' need for adequate water supply for agricultural production, the audit team evaluated the NIA's performance in delivering its mandate. A key indicator used in the assessment was the percentage of irrigated areas vis-à-vis the estimated total irrigable area. As such, the reliability of the estimated total irrigable area is fundamental to a fair assessment of performance. Additionally, alignment of this estimates with the National Irrigation Master Plan (NIMP) 2020–2030 was also evaluated, discussed in the subsequent paragraphs.
5. The NIMP 2020–2030 serves as the Government's primary long-term strategic framework for the planning, development, management, and modernization of irrigation systems in the Philippines in support of national objectives on food security, agricultural productivity, rural development, and poverty reduction. Prepared by the University of the Philippines–Los Baños (UPLB) and commissioned by the National Irrigation Administration (NIA) in coordination with the National Economic and Development Authority (NEDA) now Department of Economy, Planning and Development (DEPDev), the NIMP updated the previous irrigation master plan by incorporating improved analytical methodologies, updated geospatial data, revised estimates of potential irrigable areas, and policy directions arising from recent laws and national development strategies, including the Rice Tariffication Act, the Free Irrigation Service Act, and the Philippine Development Plan.
6. To operationalize its strategic direction, the NIMP established one baseline scenario and three development scenarios with corresponding physical targets and investment requirements. The Baseline Scenario reflects existing irrigation development practices with a minimum acceptable annual growth rate of 2.5 percent, while Scenario 1—identified by NIA as the primary planning reference—assumes a national rice security policy targeting 98 percent rice self-sufficiency and an annual irrigated area growth of 2.8 percent, resulting in a projected total irrigated area of about 2.60 million hectares by 2030. Scenarios 2 and 3 represent more aggressive expansion paths. Under both the



Baseline Scenario and Scenario 1, the NIMP sets annual and cumulative targets for Projected Operational Firm-Up Service Area (FUSA), which serve as the official benchmarks for assessing irrigation development performance. Presented hereunder is the comparison of the irrigated area projections under Scenario 1 compared with the Estimated Total Irrigable Area as per *Status of Irrigation Development – Based on Inventory* in Table 2.

Table 2 – Comparison of irrigated area projections under Scenario 1 and Estimated Total Irrigable Area as per Status of Irrigation Development – Based on Inventory

Year	Per Scenario 1	Per Status of Irrigation Development – Based on Inventory
	Projected Operational FUSA (ha)	Estimated Total Irrigable Area (ha)
2020	1,974,339	3,128,631
2021	2,029,620	3,128,631
2022	2,086,450	3,128,631
2023	2,144,870	3,128,631
2024	2,204,927	3,128,631
2025	2,266,665	3,128,631
2026	2,330,131	n/a
2027	2,395,375	n/a
2028	2,462,445	n/a
2029	2,531,394	n/a
2030	2,602,273	n/a

7. Verification disclosed that NIA consistently use the estimated total irrigable area of 3,128,631.00 hectares from CYs 2016 to 2025, based on a three percent (3%) slope criterion, as reflected in its reports. This figure, however, is not aligned with the NIMP’s projected Operational FUSA, which indicates significantly lower total irrigated area requirements by 2030—2,602,273 hectares under Scenario 1. Consequently, NIA’s reported performance has been assessed against a planning baseline substantially different from the physical targets established under the NIMP.

8. The Corporate Planning Division (CorPlan) explained that the estimated irrigable areas used for internal monitoring were not based on the NIMP, as the Management considered the Master Plan figures no longer reflective of prevailing conditions of the irrigable areas. Audit inquiry further revealed that the basis of the 3,128,631-hectare estimate could not be reliably substantiated due to the absence of verifiable documentation or technical justification. Even long-tenured personnel could not provide definitive records or references to support this figure, despite its continued use over time.

9. The CorPlan also noted hesitancy to revised the estimate to align with the NIMP citing concerns that changes could be questioned by guiding and oversight agencies. While they asserted that certain NIMP assumptions were outdated, no formal multi-agency technical review—including oversight and planning bodies such as DEPDev, Department of Agriculture (DA), and UPLB, was documented. Although coordination with technical partners and government agencies occurred in planning activities, it was not institutionalized into a structured process for validating or revising national benchmarks. As a result, internal reference figures were adjusted independently, without a harmonized national framework.



10. Nonetheless, CorPlan indicated that the reported figure of 3,128,631 hectares is subject to revision upon the approval of the Regional Irrigation Master Plan (RIMP), which may serve as a more credible baseline for future target-setting.
11. In view of the absence of a clearly established and documented basis for NIA's estimated total irrigable area, combined with inconsistencies between its internal benchmarks and NIMP targets, and pending the approval of the RIMP, accurate performance assessment has been hindered and the effectiveness of the NIMP as a planning and monitoring tool weakened. Consequently, the reliability of the reported figure cannot be established, undermining the credibility of irrigation development data used for performance evaluation and public reporting.
12. **Moving forward, we recommend the Management to:**
 - a. **Review and update irrigation development targets through the approval and adoption of the RIMP, based on current implementation capacity, funding availability, climate and watershed conditions, land use changes, and updated geospatial and inventory data, to amend current estimates to reflect more reliable figures that is supported with technical references;**
 - b. **Strengthen documentation and institutional knowledge management by preserving key technical assumptions and planning data through centralized records, standardized reporting templates, and proper turnover of planning and inventory information; and**
 - c. **Coordinate with oversight and planning agencies on updates to the NIMP, such as DepDev, DA and UPLB, to validate updated irrigable area estimates, assess the continued relevance of NIMP assumptions, and determine whether a formal update or supplementation of the NIMP is warranted.**
13. May we have your comments on the foregoing audit observations within fifteen (15) calendar days from receipt hereof.


COA Signed
2026-01-13
17:43:57
FRANIROSE U. MALONG
OIC-Audit Team Leader
Audit Team 1 - Corporate Fund

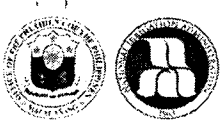

COA Signed
2026-01-13
17:45:12
JOHN GILBERT P. PIO
OIC - Supervising Auditor
Audit Group B - NIA



Proof of Receipt of AOM:

Name / Position	Received by	
	Printed Name / Signature	Date
Engr. EDUARDO EDDIE G. GUILLEN Administrator National Irrigation Administration (NIA) EDSA, Quezon City		
Engr. ROBERT C. SUGUITAN Senior Deputy Administrator Office of the Senior Deputy Administrator		
Engr. REYNE B. UGAY Officer-in-Charge Office of the Deputy Administrator for Engineering and Operations Sector		
Engr. FERDINAND G. SALES Acting Department Manager Office of the Corporate Planning Services		
Engr. PATRICK R. RILLON Acting Division Manager Planning and Programming Division		





Republic of the Philippines
OFFICE OF THE PRESIDENT
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE

January 19, 2026

MR. JOHN GILBERT P. PIO
OIC – Supervising Auditor
Audit Group B – National Irrigation Administration
Commission on Audit

Attention: **MS. FRANIROSE U. MALONG**
OIC – Audit Team Leader
Audit Team 1 – Corporate Fund

SUBJECT: Submission of Response on the Status of National Irrigation Master Plan (NIMP) 2020–2030 Recommendations and Clarification of the Estimated Total Irrigable Area

Dear **Mr. Pio**:

In reference to your communication dated *January 6, 2026*, and our subsequent meeting regarding the National Irrigation Master Plan (NIMP) 2020–2030, please find attached the National Irrigation Administration's (NIA) response on the status of each recommendation listed in the NIMP 2020–2030, as requested (**Annex A**).

Furthermore, with respect to the concern on the Estimated Total Irrigable Area of 3,128,631 hectares reported in the NIA Annual Published Report on the Status of Irrigation Development, we wish to clarify that the maps supporting this figure are not available. In the attached response, we have provided additional context and historical references pertaining to this figure for your consideration (**Annex B**).

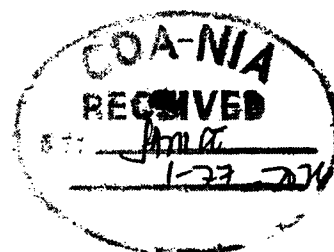
We trust that the attached documents sufficiently address your request. Should further clarification or supporting documentation be required, we remain at your disposal.

We express our appreciation to COA for its continued guidance and cooperation in ensuring the integrity and transparency of irrigation planning and reporting. Thank you.

Sincerely,

ENGR. EDUARDO EDDIE G. GUILLEN
Administrator

RCS-01
RBU-
FGS-
PRR-
RDL-
IGDA-





Republic of the Philippines
OFFICE OF THE PRESIDENT
NATIONAL IRRIGATION ADMINISTRATION
CENTRAL OFFICE



January 19, 2026

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REPUBLIC OF THE PHILIPPINES
COMMISSION ON AUDIT
 Corporate Government Audit Sector
 Cluster 5 – Agricultural and Natural Resources

Office of the Auditor
Audit Group B – National Irrigation Administration

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
<u>1. Establishment of a National Geodatabase</u> It is recommended that a comprehensive geodatabase be created to integrate both national and regional geospatial datasets. This geodatabase shall include standardized data content and nomenclature, well-defined attribute design and structure, the use of attribute domains, and the incorporation of the GIS-based Assessment, Monitoring and Evaluation (GAME) model to support evidence-based irrigation planning and decision-making.	Yes	From CY 2020 to CY 2024, the National Irrigation Administration (NIA) progressively established and strengthened its geospatial information system in support of irrigation planning and implementation under the National Irrigation Master Plan (NIMP) 2020–2030. At the Central Office (CO), one (1) ArcGIS 10.5 Advanced Desktop License was installed, and GIS workstations for mapping on-going, proposed, and existing irrigation systems were established. Standard GIS workstations and software for parcellary mapping were operationalized across all seventeen (17) regions and fifty-three (53) Irrigation Management Offices (IMOs), meeting the minimum spatial data processing requirements at national and sub-national levels. By CY 2021, NIA successfully consolidated national geospatial datasets at the Central Office, including data for 111 on-going and proposed irrigation projects and 88 validated National Irrigation System (NIS) parcellary maps. These datasets served as the foundation for the developing national geodatabase and supported evidence-based planning and project validation. By CY 2022, the geodatabase system and associated GIS infrastructure were already established and fully operational across offices. In support of governance and coordination, NIA issued <i>Memorandum Circular No. 2021-076</i> dated 04 June 2021, creating a Steering Committee and Technical Working Group for Geodatabase Development to ensure alignment with the implementation of the NIMP 2020–2030.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Further strengthening of ICT and GIS infrastructure was undertaken in CY 2023 through the acquisition of fifteen (15) Lenovo laptops, twenty-five (25) ThinkStation desktops, eight (8) Epson printers, and three (3) FGIS servers to support data processing, storage, and spatial analysis requirements. These investments enhanced system reliability, data sharing, and inter-office coordination. Unmanned Aerial System (UAS) equipment for field validation and data processing was likewise established across regions and IMOs, supporting ground-truthing activities, parcellary validation, and spatial accuracy improvements in irrigation mapping. Capacity-building initiatives were continuously implemented. In CY 2020, three (3) regions participated in GIS-related training programs, followed by four (4) regions in CY 2021. By CY 2022, all seventeen (17) regions participated in training activities focused on potential irrigable area mapping. In CY 2024, sixteen (16) regions were involved in the updating of potential irrigable area maps, ensuring data currency and consistency with planning requirements. Significant progress was also achieved in the digitization of parcellary maps. For NIS, twenty-seven (27) systems were completed in CY 2020, forty-two (42) in CY 2021, and sixty-nine (69) in CY 2022, resulting in a total of one hundred twenty (120) NIS parcellary maps validated. Additional digitization activities continued in CY 2023 with thirty (30) NIS completed and in CY 2024 with five (5) NIS completed. For Communal Irrigation Systems (CIS), service area delineation was completed for three hundred eighty-eight (388) systems, with full CIS parcellary mapping programmed after the completion of NIS parcellary mapping. In terms of human resource augmentation, NIA hired one (1) Senior Management Information/Systems Analyst at the Central Office responsible for the establishment and management of the geodatabase. Existing GIS personnel include one (1) mapper for on-going and proposed projects and three (3) mappers for existing NIS parcellary mapping at the CO. At the field level, two (2) personnel each at the RIOs (17 regions) and IMOs (53 offices), or a total of approximately one hundred forty (140) personnel, were designated for NIS parcellary digitization. In CY 2023, additional staffing was provided through the hiring of seventeen (17) Cartographer A positions and seventeen (17) Data Encoder positions, one per region, to further institutionalize GIS operations.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		However, full compliance with the recommended incorporation of the GIS-based Assessment, Monitoring and Evaluation (GAME) model has not yet been achieved. While NIA already maintains an operational GIS environment and standardized geospatial datasets, the GAME model requires a more advanced and integrated analytical platform, including high-performance computing infrastructure, specialized software, and a dedicated technical unit with advanced modeling expertise. At present, NIA's existing manpower complement, hardware capacity, and licensed software are sufficient for GIS mapping, database management, and spatial analysis, but are not yet adequate to fully operationalize the GAME model in its intended complexity and scale. In view of these constraints, NIA has adopted a phased and pragmatic approach by maximizing existing GIS resources and personnel while laying the groundwork for future GAME model adoption, subject to the availability of additional budgetary support, ICT infrastructure, and specialized human resources. Overall, the status of implementation may be considered partial to substantial compliance, with the core geodatabase infrastructure and institutional mechanisms already established, and the remaining gaps, particularly the GAME model clearly identified for future enhancement under subsequent investment programming and organizational strengthening initiatives.
<u>2. Mainstreaming Climate Change Adaptation in Irrigation Development</u> It is recommended that climate change adaptation and appropriate technologies be fully integrated into irrigation development and management. These shall include watershed protection and management, modernization of irrigation systems, integration of climate change risk assessment into Environmental and Social Impact Assessments (ESIA), water supply augmentation, water conservation and storage, and the utilization of alternative water sources.	Yes	The mainstreaming of climate change adaptation in irrigation development under the NIMP is demonstrated through sustained policy implementation, programmatic investments, and measurable physical accomplishments undertaken by the National Irrigation Administration through its Climate Change Adaptation Works (CCAW) and irrigation modernization initiatives. At the policy and institutional level, the issuance and enforcement of <i>Memorandum Circular No. 42, Series of 2022, as strengthened by Memorandum Circular No. 21, Series of 2023</i>, institutionalized the integration of climate change considerations into irrigation planning, design, construction, operation, and monitoring. These policy issuances provide an operational framework that embeds climate risk screening, system readiness assessment, and phased modernization into irrigation development, consistent with the adaptation strategies articulated in the NIMP. At the programmatic level, the NIA has implemented nationwide irrigation modernization and climate adaptation interventions, including dam upgrading, retrofitting, and rehabilitation; establishment of dam automation systems; procurement and installation of automated intake and sluice gates;

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		installation of telemetered rainfall and river gauges and other monitoring sensors; and development of dam information systems and flood forecasting and warning mechanisms. These interventions directly address climate-related risks by improving water regulation, operational efficiency, real-time monitoring, and early warning capacity. Concrete accomplishments further demonstrate the operationalization of these initiatives. The Dam Automation System installed at Malasila Dam in Barangay New Barbaza, Mlang, Cotabato was completed and commissioned on 23 August 2023, enhancing discharge control accuracy and flood responsiveness. This was complemented by the upgrading and operationalization of the NIA Command Center on 1 December 2023 at the Central Office, which now serves as the agency's central monitoring, coordination, and communications hub for dam operations, flood forecasting, and emergency response. The Command Center has since been utilized for inter-agency coordination and disaster preparedness activities. On 17 September 2025, the NIA coordinated with the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) and the Korea International Cooperation Agency (KOICA) to strengthen collaboration on flood forecasting and disaster preparedness, including discussions on the application of an integrated digital platform for complex disaster response at major dams. This was followed by an inter-agency knowledge-sharing forum on 21 November 2025 involving the Department of Education (DepEd), Department of the Interior and Local Government (DILG), Department of Public Works and Highways (DPWH), Department of Transportation (DOTr), PAGASA, Office of Civil Defense (OCD), Armed Forces of the Philippines (AFP), Philippine National Police (PNP), and Project Nationwide Operational Assessment of Hazards, which highlighted the NIA Dam Information Management System as a tool for multi-sectoral risk reduction and resilience. Operational readiness and coordination were further demonstrated during typhoons, when the Agency activated its Command Center for real-time reporting by field offices and coordinated dam operations and response measures with the OCD and the National Disaster Risk Reduction and Management Council (NDRRMC). From an investment perspective, CCAW have been implemented on a sustained, multi-year basis. In CY 2020, an allocation of ₱998.712 million covered 127 national irrigation systems. In CY 2021, ₱1.1

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		billion was allocated to 169 national irrigation systems for interventions including watershed management, renewable energy-powered irrigation, and telemetered monitoring systems. In CY 2022, allocations increased to ₱2.064 billion across 108 national irrigation systems. For CY 2023, ₱2.632 billion supported 66 national irrigation systems, including dam retrofitting, automation, telemetry installation, watershed management, and emergency communication systems. In CY 2024, ₱1.875 billion was allocated to 20 projects nationwide, achieving 93.87 percent physical accomplishment and 64.28 percent financial utilization as of December 2024. Overall, these results indicate that climate change adaptation has been integrated into the planning, implementation, monitoring, and operational management of irrigation systems through concrete infrastructure investments, institutional mechanisms, and inter-agency coordination. While full nationwide coverage remains subject to system readiness and funding availability, the actions undertaken are aligned with and supportive of the strategic objectives of the NIMP.
3.Strengthening Disaster Risk Reduction and Management (DRRM) It is recommended that irrigation planning incorporate disaster risk reduction and management options, including flood and drought forecasting systems, early warning systems for floods, landslides, and mudflows, as well as drought forecasting systems to enhance resilience against climate-induced hazards.	Yes	In planning for irrigation development, the mitigation of potential hazards such as landslides and floods is addressed through both vegetative and structural countermeasures. These measures, which include the use of fire-resilient indigenous species, wattling, gabions, and riprap, are thoroughly analyzed and integrated into the Watershed Management Plan, Land Acquisition and Resettlement Plans, and their implementation. Furthermore, buffer zones are established during the construction and operational phases to serve as fire lines, protect biodiversity, and ensure public safety. Field monitoring and evaluation, supported by research and environmental models focusing on the temporal analysis of land cover and sedimentation, are conducted to track hazard trends. This data is utilized to enhance planning and implementation strategies, ultimately reducing disaster risks within the irrigation system. The Agency has mainstreamed climate risk assessment and disaster risk reduction and management (DRRM) principles into irrigation planning by integrating adaptive measures aimed at reducing climate-induced hazards, exposure, and vulnerability. These measures include adjustment of the Cropping Calendar and Pattern of Planting (CCPP), establishment of Early Warning and Flood

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Forecasting Systems, and implementation of Dam Operation Protocols. The proposed Memorandum of Understanding (MOU) between the National Irrigation Administration and the Philippine Space Agency integrates the use of satellite data for flood forecasting using flood models to improve the accuracy of dam releases and minimize downstream flooding risks. The integration of flood forecasting models to support the automation of dam operations highlights the significance of flood forecasting models in supporting the major enhancement goal of Dam Automation, as outlined in MC 42 s. 2022. As a climate adaptation measure, the adjustment of the Cropping Calendar and Pattern of Planting (CCPP), pursuant to <i>Memorandum Circular No. 10, Series of 2021</i>, addresses projected shifts in rainfall patterns and the increasing frequency and intensity of typhoons attributed to climate change. This intervention aims to reduce exposure of agricultural production and irrigation assets to weather and flood hazards, thereby minimizing climate-related losses and damages. The establishment and operationalization of Early Warning Systems and Flood Forecasting Systems are prioritized under Climate Change Adaptation Works (CCAW), in accordance with <i>Memorandum Circular No. 42, Series of 2022</i>. The technical design standards, system specifications, and implementation guidelines are further prescribed in <i>Memorandum Circular Nos. 167, Series of 2020; 19, Series of 2023; and 13, Series of 2024</i>. Dam Protocols are enforced during typhoon events to manage reservoir levels, regulate discharge operations, and prevent structural failure. These protocols are designed to reduce downstream flood risk, protect critical irrigation infrastructure, and safeguard lives and livelihoods, particularly in climate-vulnerable downstream communities. In addition, the NIA Modernization Program includes the inclusion of forecasting systems especially for flood-related disaster, installation of automatic weather systems, and stream gaging apparatus to

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		strengthen the watershed modeling efforts of NIA to develop working models to be used as reference for extreme events simulations and climate change impacts on the watershed and service areas. Further, following the revised Environmental Impact Assessment process and outline proposed by the Environmental Management Bureau includes the Climate Change Adaptation and Disaster Risk Reduction as new chapters in the Environmental Impact Statement.
<u>4.Development of Climate- and Disaster-Resilient Engineering Standards</u> It is recommended that new engineering design standards be developed and enforced, together with strict quality assurance and quality control mechanisms during project construction, to ensure climate and disaster resilience of irrigation infrastructure.	Yes	NIA has developed a program for the sustainability of sub-watersheds called Rehabilitation and Protection of Water Resources Supporting Irrigation Systems (RPWRSIS). Through this program, vegetation and structural measures were designed based on the biophysical structures and needs for ecosystem sustainability balancing the environment, economic, and social/institutional aspects. Under the proposed Memorandum of Understanding (MOU) between the NIA and the Philippine Space Agency, the use of satellite imagery for post-processing will benefit the irrigation planning for site suitability and design through precision engineering in terms of accurate calculation of canal areas, routing, and dam/reservoir capacity. Additionally, the application of satellite imagery in irrigation demand forecasting enables NIA to assess crop water stress and implement equitable water rationing during the dry season. Included in the Feasibility Studies, the new dams designed and modernization activities are analyzed and designed not only for the agricultural sector, but also to detail the impacts of the project to the environment and vice versa, and considered in the structure planning design. The project incorporates comprehensive mitigation measures—including watershed sustainability, social safeguards, and environmental compensation—aligned with the Environmental Management Plan (EMP) and Land Acquisition and Resettlement Frameworks/Plan (LARF/Plan). The preparation for dam automation designs shall be undertaken only after the conduct and documentation of a readiness assessment confirming that an existing National Irrigation System (NIS)

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		qualifies for modernization interventions intended to enhance climate resilience. The selection of NIS for inclusion under Climate Change Adaptation Works (CCAW) projects shall strictly comply with the eligibility criteria prescribed under Section III, paragraph 2 of <i>Memorandum Circular No. 42, Series of 2022</i>. Furthermore, to ensure compliance with safety and resilience standards and to protect the structural integrity of NIA dams against climate-related risks, the Agency has implemented upgrading/retrofitting and strengthening/rehabilitation works in accordance with <i>Memorandum Circular No. 21, Series of 2023</i>.
<u>5. Irrigation Support for Crop Diversification</u> It is recommended that irrigation development explicitly support crop diversification through the identification of diversified cropping options and appropriate cropping patterns suited to the country's various climatic zones.	Yes	The Agency has substantially complied with the NIMP 2020–2030 recommendation to explicitly support crop diversification through irrigation development, including the identification of diversified cropping options and cropping patterns suited to the country's various climatic zones. Even prior to the NIMP period, NIA institutionalized crop diversification through Memorandum Circular (MC) No. 63, Series of 2019, "Adoption of Diversified Cropping in National/Communal Irrigation Systems," which provided the foundational policy framework for encouraging irrigators' associations and farmers to adopt diversified and high-value crops beyond mono-cropping, consistent with local agro-climatic conditions. To directly operationalize the NIMP recommendation and strengthen monitoring and accountability, NIA issued MC-2022-164, "Adoption of the Online Monitoring for Diversified Cropping," which institutionalized the systematic, crop-specific monitoring of diversified cropping areas. This was further reinforced by MC No. 146, Series of 2022 (23 December 2022), establishing the Guidelines for the Collection, Updating, and Validation of Irrigation Data through the Online Inventory of Irrigation Systems (OIS). The OIS captures data required under the NIMP Results Matrix, including irrigated areas planted to sugarcane, coconut, banana, corn, pineapple, mango, tobacco, and other high-value crops, thereby directly supporting evidence-based reporting and performance assessment. Within this policy and monitoring framework, the Gulayamanan / Gulayan sa Canal Program, institutionalized through MC No. 87, Series of 2024, serves as a practical, community-level

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		mechanism for advancing crop diversification and food security. Anchored on Executive Order No. 101 and the government's Enhanced Partnership Against Hunger and Poverty (EPAHP), the program promotes the productive use of canal embankments, ridges, and auxiliary spaces—consistent with MC No. 55, Series of 2014, which allows the planting of non-deep-rooted crops such as vegetables, subject to operational safeguards and prior approval by the Irrigation Management Office (IMO). Under this program, irrigators' associations cultivate a wide range of vegetables and short-gestating crops—such as pechay, eggplant, okra, tomato, mungbean, squash, and other leafy and fruit vegetables—thereby directly contributing to diversified cropping systems while simultaneously supporting canal maintenance, reducing illegal dumping, and preventing obstructions along irrigation easements. The program also complements KADIWA sa NIA, strengthening farm-to-consumer market linkages and enabling farmers to generate additional income while providing affordable produce to local communities. Building on these gains, NIA further institutionalized and scaled up the initiative through MC-2025-010, "Agency-Wide Implementation of the 'Gulayan sa Patubigan' Program: Promoting Community Gardening for Sustainable Agriculture." This Circular mandates nationwide implementation across all Regional Offices, covering both National Irrigation Systems (NIS) and Communal Irrigation Systems (CIS), and provides detailed operational procedures spanning preparation, implementation, culmination, marketing, funding, and monitoring and evaluation. The program explicitly aligns planting and harvesting cycles with canal maintenance activities and encourages partnerships with LGUs, the Department of Agriculture, seed companies, and market institutions to ensure sustainability and impact. As of 31 December 2025, the Gulayamanan sa Patubigan Program has engaged 565 Irrigators' Associations, benefiting 36,417 farmer-members nationwide. Regional implementations—including early adopters such as the Alip River Irrigation System in Maguindanao del Sur and Region III's seed-assisted canal-side gardening initiatives—demonstrate the program's adaptability across diverse climatic zones and irrigation contexts.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Under Output F: Realigned irrigation development to support diversified agricultural production systems, NIA recorded the following accomplishments aligned with the indicator “Increased irrigated area for diversified crops”: • CY 2020 (as of December 31, 2020) Wet: 41,521.86 ha Dry: 74,804.37 ha The top crops planted are corn, mungbean, okra, tomato, eggplant, winged beans, squash, upo, watermelon, peanut, cassava, pineapple, bananas, and other green leafy vegetables in MARIIS, UPRIS, and Regions I, IV-B, VII, X, XI, and XII which has the largest planted with diversified crops. • CY 2021 (as of December 31, 2021) Wet: 65,703.33 ha Dry: 59,768.64 ha The crops planted are corn, mung bean, watermelon, eggplant, sweet potato, okra, string beans, squash, and ampalaya. Beginning CY 2022, monitoring shifted to crop-specific reporting, reflecting more targeted irrigation support for diversified cropping: • CY 2022 (ha): Sugarcane 10,540.27; Coconut 2,854.89; Banana 38,519.97; Corn 49,324.15; Pineapple 4,129.77; Mango 496.84; Tobacco 534.04 • CY 2023 (ha): Sugarcane 2,695.50; Coconut 407.00; Banana 7,088.39; Corn 14,328.42; Pineapple 1,472.12; Mango 77.50; Tobacco 69.30 • CY 2024 (ha): Sugarcane 3,878.20; Coconut 1,214.61; Banana 23,245.99; Corn 53,957.97; Pineapple 1,019.34; Mango 319.00; Tobacco 2,394.72 • CY 2025 (ha): Sugarcane 1,561.02; Coconut 1,581.06; Banana 22,636.55; Corn 52,889.50; Pineapple 2,693.66; Mango 630.78; Tobacco 3,515.43

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Completed (Yes/No)	Status/Remarks for Non-compliance, if any
		These results demonstrate that irrigation development and system operation have been intentionally aligned to support diversified and high-value crops, consistent with NIMP targets for increasing irrigated areas for diversified agricultural production. In parallel, NIA actively promoted farm-level adoption through Information, Education, and Communication (IEC) initiatives, particularly the “Yayamaming Pananim” materials distributed to farmers. These IECs explain the objectives and benefits of diversified cropping, such as increased cropping intensity in water-limited areas, improved food supply, higher and more stable farm income, efficient land and water management, and climate resilience, and provide practical guidance on selecting suitable high-demand non-rice crops based on market conditions, local suitability, income potential, and cropping season. This reinforces irrigation interventions by translating policy and infrastructure support into informed on-farm decision-making. Nevertheless, while irrigation systems are designed and managed to accommodate diversified cropping, actual crop selection remains influenced by non-irrigation factors such as soil characteristics, climate variability, market access, availability of farm inputs and extension services, and farmers’ preferences and risk considerations. Variations in crop-specific irrigated areas across years therefore reflect these external constraints rather than gaps in irrigation policy or system readiness. Overall, NIA’s issuance of enabling policies (MC-2019-063, MC-2022-164, and MC-2022-146), establishment of online monitoring and validated data systems, documented irrigated area accomplishments, and sustained IEC efforts collectively confirm that irrigation support for crop diversification under the NIMP has been substantially implemented, with remaining limitations attributable primarily to factors beyond irrigation control.
6.Promotion of Diverse Irrigation Technologies for Crop Diversification It is recommended that various irrigation options—such as sprinkler, drip, and furrow irrigation—be promoted to support crop diversification. Complementary irrigation support measures should include the use of shallow tube	Yes	For FY 2024, the NIA is implementing a total of 1,388 Solar Pump Irrigation Project (SPIPs) nationwide, amounting to Php18.61 billion, covering 1,264 project sites and benefiting approximately 22,728 farmer-beneficiaries. As of 31 December 2025, the program has achieved a physical accomplishment rate of 90.07% and a financial accomplishment rate of 73.63%.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	wells (STW), small water impounding systems or farm reservoirs, solar-powered irrigation systems (SPIS), pipe delivery systems, and hybrid open-channel and pipe distribution systems, as well as improved water management schemes.	7. <u>Exploration of Public-Private Partnerships (PPP)</u> Given the high cost of farm-level and pressurized irrigation systems, it is recommended that public-private partnership (PPP) arrangements be explored, particularly for irrigation development in major projects and for diversified cropping systems.
Status/Remarks for Non-compliance, if any	Completed (Yes/No)	No While no Public-Private Partnership (PPP) contract for irrigation development was implemented during CY 2020-2025, the NIA has undertaken substantial preparatory, institutional, and capacity-building actions to operationalize PPPs in the irrigation sector, consistent with the NIMP recommendation. In recognition of the high capital cost, technical complexity, and long gestation period of farm-level and pressurized irrigation systems, particularly for diversified and high-value cropping systems, NIA formally opened its doors to PPP through the signing of a Memorandum of Agreement (MOA) with the PPP Center on 29 March 2023. This institutional partnership provides NIA with technical assistance on PPP modalities, project structuring, capacity-building, and value-for-money assessment to ensure that irrigation PPPs are technically sound, financially viable, and aligned with public interest. Following the MOA, NIA implemented concrete enabling measures, including: ● Identification of 50 potential PPP projects with seven (7) revenue streams (hydropower, solar, wind, aquaculture, bulk water supply, water treatment, and carbon credits); ● Conduct of PPP Project Concept Note (PCN) development workshops in July 2023, resulting in prioritized candidate projects using multi-criteria analysis; ● Issuance of NIA MC No. 89, s. 2023, creating an Ad-Hoc NIA PPP Project Team to institutionalize PPP development within the Agency; and ● Engagement with the Asian Development Bank (ADB) through a Transaction Advisory Services (TAS) Agreement signed on 18 September 2024, which provides no-cost technical assistance for the preparation, structuring, and competitive procurement of PPP projects in major irrigation facilities such as Pantabangan Reservoir and Jalaor River Multipurpose Project Stage II (JRMPII), including floating solar systems and bulk water supply.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Completed (Yes/No)	Status/Remarks for Non-compliance, if any
		The absence of executed PPP contracts during the review period is attributed to the necessary sequencing of policy, institutional, and technical prerequisites, including feasibility studies, financial modeling, risk allocation, and compliance with the PPP project lifecycle under national guidelines. Given the strategic importance of irrigation assets, NIA adopted a risk-averse and evidence-based approach to safeguard farmer welfare, food security objectives, and government interests. Overall, while full operational compliance has not yet been achieved, NIA's actions demonstrate deliberate and progressive alignment with the NIMP strategic direction. The Agency is now positioned to advance toward actual PPP procurement and implementation in the succeeding plan period (CY 2026 onwards), consistent with the Build Better More agenda and national food security goals.
8.Strengthening Environmental and Social Safeguards It is recommended that robust environmental and social safeguard mechanisms be institutionalized to ensure that irrigation development is sustainable, environmentally sound, and socially acceptable. Priority concerns to be addressed include watershed degradation, land conversion, increasing extreme weather events, indigenous peoples' concerns, conflicting water-use laws, and the presence of illegal settlers within irrigation project areas.	Yes	The Agency has adopted and enforced environmental and social safeguard mechanisms to ensure that irrigation development remains environmentally sustainable, socially acceptable, and compliant with national policies and regulatory requirements. These safeguards are primarily institutionalized through policy issuances, inter-agency coordination, and project-level compliance mechanisms mandated by oversight agencies and national environmental and social protection laws. In line with national government requirements and the NIMP strategic direction, NIA issued Memorandum Circular No. 002, Series of 2020, prescribing the Terms of Reference (TOR) for the Conduct of Social and Environmental Impact Assessment (SEIA) for proposed National Irrigation Projects (NIPs) and Impounding Projects (IPs). The SEIA process serves as a core safeguard mechanism to identify, assess, and mitigate potential environmental and social risks associated with irrigation infrastructure development, including impacts related to land use, communities, ecosystems, and climate-related hazards. The conduct of SEIA and related safeguard measures is consistent with national environmental and social protection frameworks, including but not limited to the Philippine Environmental Impact Statement (EIS) System (Presidential Decree No. 1586), the Philippine Clean Water Act (Republic Act No. 9275), the Philippine Clean Air Act (Republic Act No. 8749), the Ecological Solid Waste Management Act (Republic Act No. 9003), and other applicable laws and regulations governing infrastructure development and natural resource management. Compliance with these policies is required by oversight agencies such as the Department of Economic Planning and Development

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Completed (Yes/No)	Status/Remarks for Non-compliance, if any
		(DEPDev), formerly NEDA, as well as funding and regulatory institutions, thereby embedding environmental and social safeguards into NIA's project development and implementation processes. Watershed protection constitutes a central component of NIA's environmental safeguard measures under the NIMP 2020-2030. Consistent with the Plan's objective of ensuring dependable irrigation water supply while minimizing flood risks, sedimentation, and watershed degradation, NIA has implemented watershed management initiatives under the Rehabilitation and Protection of Water Resources Supporting Irrigation Systems. To operationalize this mandate, NIA and the Department of Environment and Natural Resources (DENR) entered into a Memorandum of Agreement in August 2019 covering the protection and management of critical watersheds supporting National Irrigation Systems. This MOA has been continuously implemented during the NIMP period and provides the framework for inter-agency collaboration in watershed conservation, preparation of Integrated Watershed Management Plans (IWMPs), environmental permitting, and enforcement of forestry and environmental laws. Watershed management plans and assessment studies developed under this framework serve as the scientific basis for formulating strategies, programs, and interventions that enhance the resilience and sustainability of downstream irrigation systems. NIA has integrated social safeguards for Indigenous Cultural Communities/Indigenous Peoples (ICCs/IPs) in compliance with national laws and policies, including coordination with the National Commission on Indigenous Peoples (NCIP). For irrigation projects affecting ancestral lands, NIA complies with the requirement for appropriate certifications, including the Certificate of Ancestral Land Title (CALT) and other NCIP-issued clearances. This commitment is demonstrated by formal agreements such as the Memorandum of Agreement signed on 27 September 2023 among NIA, the Bugkalot Ancestral Domain Environmental Management Council (BADEMC), and the NCIP, covering Corporate Social Responsibility initiatives for the Bugkalot/Hongot Indigenous Cultural Communities. These agreements institutionalize participatory engagement, respect for ancestral domain rights, and culturally appropriate development interventions within irrigation project areas.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		To address concerns related to competing water uses and conflicting water-use regulations, NIA continues to strengthen coordination with the National Water Resources Board (NWRB) and other concerned agencies. This coordination supports the harmonization of irrigation water requirements with domestic, industrial, environmental, and other sectoral demands, consistent with integrated water resource management principles emphasized under the NIMP. While environmental and social safeguard mechanisms are implemented through policy mandates, project-level procedures, and inter-agency coordination, NIA recognizes that there is currently no single, dedicated office within its existing organizational structure solely responsible for the integrated management of environmental and social safeguards across all agency operations. Safeguard functions are presently embedded within existing offices as part of compliance with regulatory requirements for infrastructure development. To address this institutional limitation and further align with the NIMP 2020-2030 targets, the strengthening and streamlining of environmental and social safeguard functions have been incorporated into NIA's proposed reorganization plan. The proposed structure aims to formalize and centralize safeguard-related functions, enhance manpower and technical capacity, and ensure a more coherent, agency-wide approach to environmental protection, social acceptability, and climate and disaster resilience in irrigation development. NIA has partially established and applied environmental and social safeguard mechanisms through mandatory impact assessments, watershed management initiatives, Indigenous Peoples' engagement, and sustained inter-agency coordination. While institutional strengthening remains necessary to further enhance effectiveness and integration, existing policies, agreements, and implementation practices demonstrate alignment with the NIMP strategic direction on strengthening environmental and social safeguards and underscore the Agency's commitment to sustainable, inclusive, and resilient irrigation development.
9. Institutional Development and Organizational Reforms within NIA It is recommended that NIA pursue institutional reforms focused on competitiveness through clearly defined	Yes	The NIA has partially complied with the institutional development and organizational reform recommendations of the NIMP 2020-2030. While the policy direction and reform framework have been adopted, the full implementation of the proposed organizational restructuring remains ongoing

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Completed (Yes/No)	Status/Remarks for Non-compliance, if any
functions and deliverables, decentralized and deconcentrated operations, a broader role for Irrigators' Associations in agribusiness development, strengthened national research and development, increased private sector investment, research-based planning, and a strong accountability system. The creation of a Quality Assurance Office and a Social and Environmental Safeguard Unit is likewise recommended, along with organizational restructuring guided by defined principles.		and subject to governance and regulatory approvals, particularly from the Governance Commission for GOCCs (GCG). Since CY 2021, NIA has been undertaking a comprehensive reorganization process through an external consultancy, recognizing that institutional strengthening is a prerequisite for effective and sustainable NIMIP implementation. The proposed reorganization is critical to the execution of the entire NIMIP, as the Plan's strategic directions require additional and specialized manpower to perform expanded functions that are not adequately covered under the current staffing structure. These include strengthened monitoring and evaluation systems, establishment of quality assurance as well geodatabase mechanisms, integration of social and environmental safeguards, enhanced performance management, and research-based planning, functions that are central to results-based public sector management and long-term irrigation sustainability. To operationalize key reform areas while awaiting full approval of the reorganization, NIA instituted interim measures beginning CY 2022, including the creation of the Asset Management Unit and the Interim Constructors' Performance Evaluation System Implementing Unit (CPES-IU), which were approved by the Executive Committee and the Administrator. These interim arrangements allowed NIA to begin addressing accountability, asset optimization, and performance monitoring requirements consistent with NIMIP objectives. Progress on the reorganization was reported to the NIA Board, and the Change Management Team approved the Draft Final Report of the consultant in December 2022, reflecting organizational readiness for transition. From CY 2023 to 2025, the reform initiative progressed to the governance approval stage. The NIA Board accepted the reorganization proposal on June 24, 2024, and authorized its submission to the GCG, which was formally completed on April 28, 2025. However, due to the insufficient number of permanent positions under the existing staffing pattern, NIA has had to rely on Casual and Contract of Service personnel to support ongoing and upcoming NIMIP-related activities, including project management, monitoring, and technical coordination. While this arrangement ensures operational continuity, it is not a substitute for the institutionalized staffing and accountability mechanisms envisioned under the NIMIP. In parallel, NIA has advanced decentralization and project-based management reforms through the proposed establishment of a Unified Project Management Office and Project Management Offices for

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		major irrigation projects such as Bayabas, Calapangan, Tumauni, and INISAP, which have already been submitted for consideration. The ongoing revision of reorganization annexes further demonstrates NIA's commitment to aligning its institutional structure with the expanded scope, complexity, and safeguard requirements of NIMP implementation. In sum, NIA's partial compliance reflects a transitional and procedural limitation rather than a policy or implementation gap. The completion and approval of the reorganization remain critical to fully enabling NIA to carry out the NIMP's recommendations, particularly in the areas of monitoring and evaluation, quality assurance, geodatabase, and social and environmental safeguards, which require dedicated units, clearly defined mandates, and adequate permanent manpower to ensure effective, accountable, and sustainable irrigation development through 2030.
10. Adoption of a Results-Based Monitoring and Evaluation (RBME) System It is recommended that an RBME system with clearly specified performance indicators be adopted to ensure the achievement of physical targets and desired outcomes. This system shall be anchored on the Results Framework developed under the NIMP.	No	The Agency currently undertakes monitoring and evaluation-related functions through several existing offices, including the Construction Management Division under the Engineering Department, operational monitoring units within the Operations Department, and planning and performance monitoring functions lodged under the Corporate Planning Services, such as the CPES-IU. These offices are responsible for tracking physical accomplishments, financial utilization, and project implementation progress, thereby supporting operational oversight and compliance monitoring. However, these functions are dispersed across offices and <i>are not consolidated under a single unit</i> with an agency-wide mandate for results-based monitoring and evaluation. In line with the government's continuing efforts to strengthen all components of the public sector management cycle, namely planning, budgeting, implementation, monitoring, and evaluation—the Department of Economic Planning and Development (DEPDev), formerly the National Economic and Development Authority (NEDA), and the Department of Budget and Management (DBM) established the National Evaluation Policy Framework (NEPF) through NEDA-DBM Joint <i>Memorandum Circular No. 2015-01</i> issued in July 2015. The NEPF provides a policy framework for the purposive and systematic conduct of evaluations in the public sector in support of good governance, transparency, accountability, and evidence-based decision-making. Consistent with this framework, and during the period when NIA was an attached agency of the Department of Agriculture (DA) in 2023, the DA, acting as the Central Office Neutral Evaluation Unit (CO-NEU), required its operating units and attached agencies to establish their respective Neutral

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Evaluation Units (OU-NEUs). In accordance with the DA Memorandum dated 16 December 2022, NIA issued <i>Memorandum Circular No. 13, Series of 2023</i>, providing for the creation of the Agency's Neutral Evaluation Unit (NEU). The NEU is intended to strengthen the agency's evaluation function by ensuring objectivity, independence, and alignment with the NEPF and results-based governance principles. Notwithstanding these policy issuances, NIA currently does not have an existing office within its approved organizational structure that is fully mandated to oversee the overall monitoring and evaluation function of the agency and to operationalize a comprehensive Results-Based Monitoring and Evaluation (RBME) system anchored on the NIMP Results Framework. As a result, there is no dedicated plantilla position, manpower complement, or regular budgetary allocation specifically assigned to carry out agency-wide RBME functions, including outcomes and impact evaluation. These institutional and resource limitations affect the full operationalization of an RBME system across the agency. To address this gap, the institutionalization of the RBME function, including compliance with the NEPF and the operationalization of the NEU, has been explicitly incorporated in NIA's proposed reorganization plan submitted to the GCG. The proposed reorganization seeks to establish a centralized and dedicated RBME mechanism within the agency's organizational structure, harmonize currently fragmented monitoring functions, and ensure the availability of appropriate manpower and resources to support results-based planning, monitoring, and evaluation. This reform is critical to strengthening the linkage between planning, budgeting, implementation, and evaluation in support of the NIMP 2020–2030. To support interim monitoring requirements, NIA has established and operationalized several online and digital monitoring systems that provide timely and standardized data on project implementation and operations. These include the Online Monitoring System for Contractors and Ongoing Contracts (<i>MC No. 108, s. 2020, and MC No. 161, s. 2020</i>); the Enhanced Physical and Financial Accomplishment Monitoring Information System (<i>MC No. 178, s. 2020</i>); the Farmland Geographic Information System (FGIS) adopted as the official source of data for operations and maintenance of irrigation systems (<i>MC No. 076, s. 2023</i>); the Online Monitoring System for Diversified Cropping (<i>MC No. 164, s. 2022</i>); the Online Inventory of Irrigation Systems (OIS) for data collection, updating, and validation (<i>MC No. 146, s. 2022</i>); and the Online Preparation of Office and Division Performance

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Commitment Review (ODPCR) System (<i>MC No. 137, s. 2024</i>). While these systems significantly enhance transparency, data consistency, and operational performance tracking, they remain largely output- and compliance-oriented and are not yet fully integrated into a unified RBME framework. Overall, while a fully institutionalized RBME system anchored on a comprehensive Results Framework has yet to be fully established, significant preparatory actions and policy reforms have been undertaken. Full realization of the RBME system under the NIMP 2020-2030 is expected upon the approval and implementation of the Agency's reorganization proposal, the full operationalization of the NEU, and the integration of existing monitoring platforms into a unified results-based monitoring and evaluation framework capable of measuring outcomes and long-term impacts of irrigation investments.
11. Formulation of Regional and Provincial Irrigation Master Plans It is recommended that regional and provincial irrigation master plans be formulated to align national strategies with local development priorities and conditions.	Yes	Consistent with the NIMP 2020-2030 recommendation on the formulation of Regional and Provincial Irrigation Master Plans, the Department of Agriculture (DA), through Special Order No. 958, series of 2022, established a Technical Working Group (TWG) to lead the preparation of the Regional Irrigation Master Plans (RIMPs) 2023-2030. The TWG is composed of representatives from the NIA, Bureau of Soils and Water Management (BSWM), and Bureau of Agricultural and Fisheries Engineering (BAFE), ensuring strong inter-agency coordination and technical consistency with national irrigation and agricultural development policies. The RIMP 2023-2030 serves as a key planning instrument that operationalizes the strategic directions, priorities, and investment frameworks of the NIMP by translating national-level policies into region- and province-specific irrigation development strategies, taking into account local agro-climatic conditions, water resource availability, physical targets, and development priorities. Through a series of extensive consultations, technical workshops, and validation meetings, the TWG, in coordination with NIA Regional Irrigation Offices, NIA Irrigation Management Offices (IMOs), and DA Regional Field Offices (DA-RFOs)—including the Regional Agricultural Engineering Division (RAED) and Planning, Monitoring, and Evaluation Division (PMED), successfully formulated and finalized all fifteen (15) RIMPs covering CY 2023-2030. Each RIMP underwent technical review and management validation and was endorsed by the Top Management of NIA, BSWM, and BAFE, signifying readiness for implementation.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Formal presentations to NIA Central Office Top Management were conducted in three (3) batches on 13 June 2024, 30 July 2024, and 11 September 2024, further affirming institutional ownership and executive oversight of the regional plans. However, the recommendation is assessed as partially complied due to the pending issuance of the signed Certificates of Completion and the approval of the Joint Memorandum Circular (JMC) between the DA and NIA. The delay in the issuance of the JMC is primarily attributable to significant institutional realignments affecting NIA's organizational supervision during the period of RIMP formulation. In 2022, pursuant to <i>Executive Order No. 168</i>, the NIA was transferred to the Department of Agriculture, aligning its operations with broader agricultural initiatives to strengthen irrigation support for food production. This organizational setup was in effect during the early and mid-stages of the RIMP formulation. Subsequently, in 2024, under <i>Executive Order No. 69</i>, NIA was transferred back to the Office of the President, reinforcing its strategic role in national food security and placing irrigation development directly under the President's supervision. As a result of this institutional transition, the proposed DA-NIA Joint Memorandum Circular underwent further legal and policy review to ensure consistency with NIA's revised mandate, administrative supervision, and signatory authority under the Office of the President. This review is necessary to safeguard the legal soundness and enforceability of the JMC, particularly with respect to implementation responsibilities, coordination mechanisms, and accountability structures. Despite the pending JMC, the substantive outputs of the recommendation have been achieved, as all fifteen (15) Regional Irrigation Master Plans have been completed, validated, and approved by the concerned agencies. The ceremonial signing of the JMC is scheduled on 13 January 2026, after which the JMC and the corresponding Certificates of Completion will be formally issued. At this stage, the remaining actions are administrative and procedural in nature, as the substantive requirements of the recommendation, namely, the formulation, validation, and approval of all fifteen (15) Regional Irrigation Master Plans have already been fully achieved. Full compliance is expected upon the ceremonial signing and formal issuance of the JMC, which will institutionalize the RIMPs as official planning and reference documents for regional and

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
12. Institutionalization of Quality Assurance, Value Engineering, and Updated Feasibility Standards It is recommended that quality control and quality assurance mechanisms be institutionalized to maintain high construction standards, and that Value Engineering/Value Analysis (VE/VA) be implemented across all stages of irrigation development. Furthermore, the minimum requirements for conducting feasibility studies should be upgraded to meet the data needs of the revised irrigation project prioritization criteria.	Yes	provincial irrigation development, investment programming, and implementation under the NIMP CY 2020–2030 framework. The Agency has substantially advanced the institutionalization of Value Engineering and Value Analysis (VE/VA) in line with the National Irrigation Master Plan (NIMP) CY 2020–2030 through the issuance and implementation of <i>Memorandum Circular (MC) No. 88, s. 2020</i>, which provides comprehensive guidelines for undertaking VE/VA during the Detailed Engineering Studies (DES) and Detailed Engineering Design (DED) stages of irrigation project development. This issuance is fully aligned with the DEPDDev (formerly NEDA) Value Analysis Handbook and reflects government-wide efforts to improve public investment efficiency and control escalating project development and construction costs. Accordingly, VE/VA has been mainstreamed as a mandatory requirement for all NIA infrastructure projects, consistent with DEPDDev investment programming and approval processes. In terms of upgrading feasibility study (FS) standards, NIA issued <i>MC No. 003, s. 2023 (Manual for the Conduct of Preliminary Feasibility Study)</i>, which standardizes and strengthens the technical, economic, environmental, and institutional analyses required at the early stages of project development. This manual enhances the quality, consistency, and reliability of feasibility data and directly supports the revised irrigation project prioritization criteria under the NIMP. To operationalize these standards, NIA institutionalized a structured and criteria-based project screening process through the conduct of the Project Prioritization Workshop for Feasibility Study and Detailed Engineering (FS&DE) for FY 2026 in Puerto Princesa City, Palawan. The workshop evaluated, validated, and prioritized proposed irrigation projects to ensure technical soundness, economic viability, and alignment with national development priorities, in strict accordance with <i>MC No. 7, s. 2021, MC No. 137, s. 2023</i>, and the respective Regional Irrigation Master Plans (RIMPs). Participation from all NIA regions and major irrigation systems enabled rigorous technical review, validation of feasibility data, assessment of project readiness and budget requirements, and compliance with established FS standards, demonstrating NIA's transition toward an evidence-based and policy-aligned investment programming process. Parallel efforts have been undertaken to strengthen quality control and technical assurance mechanisms. NIA issued <i>MC No. 186, s. 2020 (Guidelines on Materials Testing and Quality Control)</i>, which establishes standardized procedures for materials testing and construction quality control to

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		ensure the safety, durability, and performance of irrigation facilities. These policies are operationalized through continuous capacity-building and infrastructure investments, including the inauguration of the NIA Irrigation Engineering Center (IEC) Soils and Materials Testing Laboratory at the Central Office on March 24, 2022, equipped with facilities for triaxial testing, geotechnical analysis, and advanced soil testing technologies. Further strengthening institutional credibility, the IEC Chemical Laboratory received its Certificate of Authority to Operate (CATO) from the Professional Regulatory Board of Chemistry on February 27, 2024, pursuant to Republic Act No. 10657 (Professional Chemistry Law). This certification authorizes the laboratory to conduct chemical analyses of soil and water essential to irrigation water quality management and agricultural productivity, reinforcing compliance with national professional and regulatory standards. At the operational level, NIA has implemented nationwide initiatives to institutionalize quality practices, including the Planning Workshop on the Formulation of Technical Specifications for NIA Materials Testing and Quality Control held on November 28 to December 2, 2022, in Tigbauan, Iloilo, which aimed to formulate by-laws for NIA Materials Engineers, update technical specifications, and develop official manuals and standard forms. Capacity-building activities have likewise been conducted across regions, such as regionwide trainings on Materials Testing and Quality Control (e.g., NIA Region V, November 2023), advanced soil testing and geotechnical workshops (October 2023), and regional seminars on contract management, materials testing, and quality assurance (e.g., NIA MIMAROPA, September 2025). Several regional offices have also begun establishing materials testing laboratories, including the inauguration of the NIA Region IX Materials Testing Laboratory on October 23, 2024, evidencing progressive decentralization of QA capabilities. Notwithstanding these advancements, the full institutionalization of quality assurance as a distinct, agency-wide function has not yet been achieved. At present, NIA does not have a dedicated Quality Assurance Office, as envisioned under the NIMP and related reform initiatives. Limitations in organizational structure, availability of specialized technical manpower, and supporting infrastructure, particularly facilities and equipment required for comprehensive materials testing and quality assurance, continue to constrain the Agency's ability to implement a fully integrated QA system across all project stages. These challenges are further compounded by funding constraints, as the Agency's annually approved budget is largely prioritized for direct irrigation development and rehabilitation

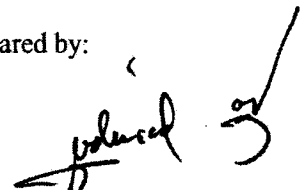
Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		activities, resulting in limited fiscal space for the establishment of new organizational units and acquisition of specialized QA resources. These constraints are therefore structural and resource-based rather than policy-related. To address these gaps, the Agency's pending reorganization proposal explicitly includes the establishment of a dedicated Quality Assurance function, together with the necessary staffing pattern and institutional mechanisms. Once approved and implemented, this reform will enable NIA to fully comply with the NIMP recommendation by institutionalizing quality assurance, fully operationalizing upgraded feasibility study standards, and ensuring the consistent application of VE/VA and QA mechanisms throughout the irrigation project lifecycle. Pending the approval of these structural reforms, the recommendation is assessed as partially complied, with strong policy foundations, demonstrable operational progress, and a clearly defined pathway toward full compliance.
13. Strengthened Inter-Agency Coordination and Policy Support It is recommended that strong coordination between NIA and the Department of Environment and Natural Resources (DENR) be institutionalized, particularly on watershed and aquifer protection and management. In addition, support should be extended to the enactment of a National Land Use Policy Act to protect agricultural lands.	Yes	The NIA has undertaken extensive inter-agency coordination and policy support initiatives to promote the sustainable management of irrigation systems, watershed protection, and agricultural development. These initiatives encompass infrastructure development, environmental management, water resource optimization, agricultural convergence programs, research and technology partnerships, and legislative and policy coordination, reflecting NIA's commitment to integrated water and land resource management consistent with the NIMP strategic directions. <i>Infrastructure, Rural Development, and Agrarian Reform Support</i> NIA has coordinated with the Department of Public Works and Highways (DPWH) and the Department of Agriculture (DA) for the construction, repair, and rehabilitation of service and access roads within irrigation systems. Service roads along canal embankments and access roads linking irrigation facilities to provincial and national road networks also serve as farm-to-market roads, directly supporting agricultural production and rural accessibility. NIA has likewise implemented the irrigation support component of Agrarian Reform Community (ARC) development in coordination with the Department of Agrarian Reform (DAR). This includes the development of new irrigated areas, rehabilitation of existing irrigation systems, and the organization and capacity-building of farmer-beneficiaries. Technical assistance has been extended to

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		local government units (LGUs) and other stakeholders planning irrigation projects. In addition, resettlement sites for families displaced by irrigation projects have been developed based on modern town-planning concepts, with provision of basic utilities and all-weather internal road networks. <i>Watershed Protection and Environmental Coordination (2019 MOA; implemented during NIMP period)</i> Watershed protection remains a core area of inter-agency coordination during the NIMP 2020–2030 implementation. The Memorandum of Agreement (MOA) between NIA and the Department of Environment and Natural Resources (DENR), signed on 05 August 2019, covering the effective management and development of 143 critical watersheds supporting National Irrigation Systems (NISs), has been continuously implemented during the NIMP period. Pursuant to DENR Memorandum Circular No. 2008-05, NIA and DENR have coordinated in the preparation of Integrated Watershed Management Plans (IWMs), identification and prioritization of watersheds requiring rehabilitation, and securing of environmental clearances and permits. NIA has also recommended qualified field personnel for deputization as Environment and Natural Resources Officers and participated in joint enforcement of forestry laws, watershed community management, information sharing, IEC campaigns, and research and development activities. Coordination meetings, including those held in January 2023, supported updates to watershed boundaries and alignment of ongoing watershed initiatives critical to irrigation sustainability. <i>Water Resource Optimization and Renewable Energy Initiatives</i> Consistent with NIMP objectives on efficient and multi-purpose utilization of water resources, NIA entered into key inter-agency agreements in 2023. A Memorandum of Agreement with the Department of Energy (DOE) was signed on 07 December 2023 to promote the use of irrigation facilities for renewable energy development. In addition, NIA, the Department of Environment and Natural Resources (DENR), the Water Resources Management Office (WRMO), and the National Water Resources Board (NWRB) signed a Memorandum of Agreement on 04 October 2023 to enable the optimal utilization of irrigation water for power generation, bulk water supply, aquaculture, recreation, and tourism, in accordance with

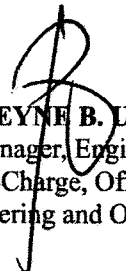
Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Executive Order No. 22. These agreements support water security and operational efficiency within the NIMP framework. <i>Agricultural Development and Convergence Programs</i> NIA has actively participated in national convergence programs during the NIMP period. The Agency signed the Joint Memorandum Circular and Addendum to the Memorandum of Understanding on the Enhanced Partnership Against Hunger and Poverty (EPAHP) on 09 December 2020, institutionalizing collaborative efforts among national agencies and partners to address food security and poverty reduction. Support to the Masagana Rice Industry Development Program (MRIIDP) was strengthened through inter-agency and private sector convergence activities, including the convergence meeting held on 30-31 May 2023, involving the DA, DPWH, and private sector partners. NIA also supported the Kapatid Angat Labat for Agriculture Program (KALAP) through partnerships with Go Negosyo and LGUs, promoting inclusive agricultural development. Coordination with the Cooperative Development Authority (CDA) resulted in the signing of a Memorandum of Understanding on 06 September 2023 for the establishment of Irrigators Service Cooperatives (ISCs), aimed at strengthening irrigators' organizations, improving farm productivity, and supporting livelihood enhancement. <i>Research, Technology, and Capacity-Building Partnerships</i> NIA expanded partnerships with academic, research, and development institutions to advance technology-driven irrigation management and climate resilience. Key milestones include the signing of a Memorandum of Understanding with DOST-PCIEERD on 27 May 2022 for the Water Resource Management Climate-Resilient Infrastructure Initiative; coordination with PhilRice and the Philippine Space Agency (PhilSA) beginning 2023 for data-driven crop monitoring and irrigation planning; and the signing of a Memorandum of Agreement with the International Rice Research Institute (IRRI) on 07 April 2025 for digital tools supporting sustainable rice farming and groundwater monitoring.

Recommended Strategic Directions and Policy Actions in the National Irrigation Master Plan CY 2020-2030	Complied (Yes/No)	Status/Remarks for Non-compliance, if any
		Additional partnerships with institutions such as the University of the Philippines Los Baños (UPLB) (November 2024) and De La Salle University (DLSU) (December 2025) further strengthened dam safety, research collaboration, and capacity-building for NIA personnel. Legislative and Policy Coordination NIA has provided technical comments and policy positions on various House and Senate bills, development plans, and proposed executive issuances related to irrigation, watershed management, water resources, infrastructure development, and agricultural productivity. This includes inputs on the revised draft Executive Order on streamlining the preparation and implementation of Comprehensive Land Use and Physical Framework Plans of LGUs last May 2024, demonstrating NIA's active role in national policy coordination. The Agency has established and maintained extensive inter-agency coordination mechanisms through formal agreements, convergence programs, joint implementation activities, and sustained policy engagement. While continued institutional strengthening and harmonization of these partnerships remain essential, the scope, depth, and continuity of coordination initiatives implemented during the NIMP period reflect strong alignment with the Plan's strategic direction on strengthened inter-agency coordination and policy support, and underscore NIA's commitment to water security, environmental sustainability, and agricultural development.

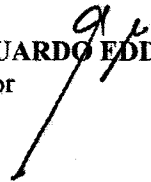
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During our meeting this morning regarding the National Irrigation Master Plan for 2020–2030, we discussed the Estimated Total Irrigable Area of 3,128,631 hectares, as reflected in the NIA Annual Published Report on the Status of Irrigation Development based on the NIA inventory.

You mentioned that the basis or source of this figure (3,128,631) could not be ascertained, as even long-term employees whom you consulted were unable to provide definitive information, and that this figure has been used consistently since your tenure in the agency.

In this regard, we respectfully request that you further expound on this matter, particularly with respect to any available documentation, historical context, or assumptions underlying the said figure.

Thank you for your continued cooperation.

Response:

As discussed during the meeting, the National Irrigation Administration (NIA) has exercised due diligence and exhausted all reasonable efforts to determine the original basis of the published **Estimated Total Potential Irrigable Area (PIA) of 3,128,631 hectares** reflected in the NIA Annual Reports and the NIA website, particularly during the formulation of the **National Irrigation Master Plan (NIMP) 2020–2030**. Available records indicate that this figure has been consistently cited since at least 2010, and possibly earlier; however, no accessible internal documentation, geospatial datasets, or formal policy issuances within NIA could be located to definitively trace its original derivation. Consultations with long-serving NIA personnel likewise yielded no verifiable historical reference.

Notwithstanding the absence of surviving internal records, the **historical technical origin of the 3.1-million-hectare figure is substantially explained in the World Bank's Philippines Irrigated Agriculture Sector Review (Report No. 9848-PH, April 15, 1992)**. According to this authoritative external review, the most frequently quoted estimate of potentially irrigable land in the Philippines—3.126 million hectares—was reported in NIA's 1990 Corporate Plan and was primarily based on the criterion of land with slopes of less than three percent (0–3%). The assessment was undertaken in the early 1980s, using 1:50,000 topographic maps dating from the 1950s, with contour intervals of 25 meters in steep areas and 5 meters in plains (World Bank, 1992, paras. 3.1–3.3).

The same World Bank report further clarifies that the delineation of this gross irrigable area was **largely slope-based and, in some cases, approximated using the extent of existing rice paddies**, which often extended across entire plain areas up to foothill zones. While certain adjustments were made for major settlements, marshes, lakes, and prominent hilly zones, **no systematic account was taken of roads, dispersed rural settlements, urban expansion, infrastructure requirements, flood hazards, drainage constraints, water availability, or the cost of delivering irrigation and flood control facilities**. Moreover, the assessment did not account for **land conversion due to rapid urbanization and population growth since the**

time of the underlying topographic mapping, and the original map sheets used to demarcate irrigable lands are no longer readily available (World Bank, 1992, para. 3.3).

The World Bank review also highlighted significant inconsistencies between national-level and provincial-level estimates of potentially irrigable land. While the national total from the 1990 Corporate Plan (3.126 M ha) and the aggregated estimates from the 1989 Provincial Irrigation Profiles (PIPs) differed by only about three percent at the national level, regional totals showed substantial divergences, indicating that these figures were derived from separate sources and methodologies rather than a single harmonized spatial database (World Bank, 1992, para. 3.4 and Table 3.1). This finding reinforces the conclusion that the historical 3.1-million-hectare figure was an aggregate planning estimate rather than a rigorously validated, spatially consistent inventory.

Consistent with these findings, further verification undertaken by NIA during the NIMP formulation showed that other government agencies, including the Department of Economy, Planning, and Development (DEPDev) and the Philippine Statistics Authority (PSA), likewise cite NIA as the original source of the 3.1-million-hectare estimate. This resulted in a circular reference, further confirming that the figure is rooted in legacy planning estimates dating back to the 1980s and early 1990s, rather than in updated or independently verifiable datasets.

In response to these well-documented limitations, the **NIMP 2020–2030 deliberately departed from exclusive reliance on the legacy 0–3 percent slope-based estimate**. The NIMP formally introduced a **more technically grounded and transparent framework** for determining irrigation potential, including the **expansion of the slope criterion to 0–8 percent**, consistent with advances in irrigation engineering, improved construction technologies, and empirical evidence that many existing national and communal irrigation systems are already operating efficiently on slopes exceeding the earlier threshold. This shift directly addresses the constraints identified in the World Bank review, which noted that the original definition considered only physical irrigability of land and did not incorporate water availability, infrastructure feasibility, or evolving land use conditions.

In support of this revised criterion, the NIMP developed and adopted **updated national maps for Gross Potential Irrigable Areas (Gross PIA)** using standardized geospatial datasets and explicit biophysical parameters, including slope and soil characteristics. These basemaps served as the primary spatial reference for irrigation potential at the national level, replacing undocumented aggregate figures with **replicable and technically defensible spatial analysis**.

Consistent with NIMP recommendations, **Regional Irrigation Master Plans (RIMPs) 2023–2030** were subsequently prepared to operationalize these national standards. The RIMPs refined the Gross PIA by incorporating additional **natural, environmental, legal, and human-induced constraints**, resulting in the delineation of **Net Potential Irrigable Areas (Net PIA)** after deducting built-up areas, roads, forestlands, mangroves, inland water bodies, fault lines, CADTs, and legislated protected areas. The RIMPs also integrated regional assessments of **socioeconomic conditions, agricultural productivity, land and water resources, water**

balance, and climate risk, thereby addressing the very limitations of the earlier slope-only methodology identified in Report No. 9848-PH.

Once the updated PIA figures from the consolidated RIMPs are approved through the **Joint Memorandum Circular (JMC) of DA–NIA**, NIA will recommend to oversight offices a **managed transition period**, during which the updated PIA will be gradually adopted while still referencing the legacy figure for monitoring continuity. This approach ensures orderly transition, avoids abrupt disruptions in planning and budgeting frameworks, and allows stakeholders to align investment and performance monitoring systems with the updated, evidence-based PIA.

In summary, the **3,128,631-hectare figure is historically traceable to early-1980s slope-based assessments documented in NIA's 1990 Corporate Plan and independently analyzed in World Bank Report No. 9848-PH (1992)**. While the original supporting datasets are no longer available, the technical assumptions, scope, and limitations of the estimate are clearly documented in said report. The **NIMP 2020–2030 was expressly designed to correct these legacy limitations** by introducing updated slope criteria, standardized geospatial basemaps, and multi-factor regional refinement, thereby providing a more reliable and auditable technical basis for irrigation planning moving forward.

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Philippines

Irrigated Agriculture Sector Review

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3. IRRIGABLE LAND AND ITS DEVELOPMENT

A. IRRIGABLE LAND RESOURCES

General

3.1 The primary definition of potentially irrigable land in the Philippines is land on a slope of less than 3%¹. The area defined by slope must then be rectified to take account of soil conditions, surface water and groundwater drainage constraints, the magnitude of flood hazards, and land permanently alienated from irrigation use by road and rail communications infrastructure and urban and rural dwellings and other buildings. Given rapid population growth and its concentration on the plains, increasing amounts of potentially irrigable land will be occupied by settlement.

3.2 Most of the land with slope of less than 3% occurs in valley bottoms of the larger rivers and on the coastal plains. The break in slope from plain to mountains is generally abrupt. Hill streams are in deeply incised, V-shaped valleys with steep gradients and with very limited level land, and there are relatively few major river systems with large associated plain lands. Much of the area is drained from small basins by rivers which flow independently to the sea. Typically, these have steep catchments and proportionately small areas of plain land. The notably extensive areas of plain lands occur in the Cagayan basin of northern Luzon, the Pampanga and Agno basins of Central Luzon, the Bicol plain of southern Luzon, the central plain of Panay Island, the north central plain of Leyte, and the plains of the Agusan, Davao and Cotabato basins in Mindanao.

Estimates of Potentially Irrigable Land

3.3 The most frequently quoted estimate of potentially irrigable land is 3.126 M ha which is given in NIA's 1990 Corporate Plan. This is said to be based on the primary criterion of land with less than 3% slope. The assessment was carried out in the early 1980s based on an analysis of 1:50,000 topographic maps -- with 25 m contours on steep slopes and 5.0 m contours on the plain lands -- dating from the 1950s. These maps indicate areas then predominantly under rice paddies. Examination of some sheets indicates that the definition of the gross irrigable area was sometimes based on the area under paddies (which often extends across the entire plain area to the foothill zone) although in other cases the land slope definition was applied. Adjustments were made within the gross area to account for major settlements and prominent marshes, lakes and hilly zones. However, no account was taken of infrastructure such as roads, nor of the many scattered areas of rural settlement and associated gardens and plantations, areas subject to heavy seasonal flooding, and areas which would have to be dedicated to irrigation, drainage and flood control infrastructure. The definition considered only irrigability of the land, and no account was taken of the availability of water or the costs of delivering it and of providing drainage and flood control facilities. The assessment has not been updated to take account of

¹ This criterion might be violated by including somewhat steeper land if it is already terraced for paddy cultivation and water could be easily delivered.

land taken over for new urban (and rural) settlement since the date of the topographic mapping. Unfortunately, most of the 1:50,000 sheets on which the lands regarded as irrigable were demarcated are now not readily available.

3.4 Another source, ostensibly more detailed than the 1990 Corporate Plan, is the 'Provincial Irrigation Profiles' (PIPs) prepared by NIA in 1989 for the 73 provinces with significant irrigation potential (only the scattered island province of Batanes in the Cagayan Region, Metro Manila, and small islets are omitted). The PIPs are in considerable detail and include estimates of irrigable area, present development in terms of service areas of national and communal irrigation schemes, and new developments categorized as national, communal, SWIM, and pump irrigation projects (Annex 1). However, although the countrywide totals given in the Corporate Plan and as derived by totalling the PIP estimates agree within about 3% (3.126 M ha versus 3.036 M ha), the regional totals have major differences for every region (see Table 3.1). These are so great as to indicate that they must be derived from separate sources.

Table 3.1: Comparison of Estimates of Potentially Irrigable Land from the 1990 Corporate Plan and the 1989 Provincial Profiles

Regions	Potentially irrigable area (ha)		Ratios of Corp. Plan to PIPs
	Corp. Plan	PIPs	
Ilocos	309,810	400,590	0.77
Cagayan Valley	539,710	460,840	1.17
Central Luzon	482,220	441,050	1.09
Southern Tagalog	263,590	238,680	1.10
Bicol	239,650	208,640	1.15
Western Visayas	197,250	171,490	1.15
Central Visayas	50,740	70,270	0.72
Eastern Visayas	84,380	109,450	0.77
Western Mindanao	76,500	70,110	1.09
Northern Mindanao	230,150	249,970	0.92
Southern Mindanao	290,250	241,520	1.20
Central Mindanao	362,080	374,230	0.96
Total	3.126.330	3.036.850	1.03

Source: NIA's Corporate Plan (1990) and PIPs (1989).

3.5 Other estimates of potential irrigable area can be derived from publications of the Bureau of Soils and Water Management (BSWM) of the Department of Agriculture. In 1985, the Bureau initiated the 'Land Resources Evaluation Project' which has produced a detailed report for each province of the land resources and present and potential uses. Perhaps easier to use for regional planning of irrigation will be the results of the Bureau's on-going program to produce a 'Framework for Crop Development and Soil Conservation Planning'. The assessment is based on administrative regions, and each report is supported by regional thematic maps at 1:250,000 scale, of which the 'Rice-Based Cropping System Development and Soil Conservation Planning Guide Map' is particularly useful. The mapping identifies areas which presently have paddy rice in the cropping pattern under seven categories (of which three are irrigated paddy); areas which do not presently include paddy rice in the cropping pattern but are recommended for conversion to paddy; and a class of

present waste lands which are recommended for reclamation for possible cultivation of paddy. As paddy is essentially the only irrigated field crop, this mapping will be very useful for defining the potentially irrigable areas on a sounder basis than slope alone.

Service Areas of Irrigation Projects

3.6 Uncertainties relating to potential irrigable area are also encountered at a detailed project level. Tables A.2 and A.3 summarize estimates of service area from successive implementation stages for a number of World Bank and ADB assisted projects. The results for the nine projects with Project Performance Audit Reports (PPARs) (with multiple loans for the same scheme considered as one project) are summarized in Table 3.2 with service area estimates also compared to actual 1989 wet season (WS) irrigated area.

Table 3.2. IBRD- and ADB-Supported Projects with PPARs: Service Area Estimates

	Service Area (ha)				WS Area 1989 (ha)
	Appraisal	PCR	PPAR	Actual 1989	
IBRD Projects (5)	283,170	274,601	241,331	234,001	202,944
ADB Projects (4)	39,420	31,590	31,387	27,920	23,608
Total (9)	322,590	306,191	272,718	261,921	226,552
% of Appraisal Estimate	100	95	85	81	70

Source: Attachment 1, Tables A.2 and A.3.

3.7 Service area figures are adjusted periodically and the differences indicated by Table 3.2 are to some extent arbitrary (that between the PCR and PPAR totals for instance was almost entirely due to adjustments made by the PPAR for MARIIS). Nevertheless, a 20% difference between actual service area and design estimates has been generally confirmed by a recent NIA review of six existing systems which involved detailed field surveys and hydrological reassessments. Furthermore, the difference between revised service area and actual wet season irrigated areas suggests if anything that 20% may be too low. NIA is currently completing a systematic 'firming' up of all service areas in national systems based on detailed cadastral mapping financed under Bank-supported projects and this should provide more definitive estimates.

3.8 Similar reductions in service area, and wet season irrigated areas significantly below expectations, have also been encountered in respect of communal irrigation schemes. As a result, improved procedures for estimating service area have been included under the recently-approved Bank-supported second communal project (CIDP II).

Reassessment of Irrigation Potential

3.9 Significant reductions in wet season irrigated areas for both NISs and CIS can have a potentially major impact on economic returns (Chapter 4). The 'firming up' of NIS areas and revised procedures for CIS indicate that NIA recognizes this problem. To complement these project-level activities, an

attempt has been made within the framework of the IASR to develop a more rigorous evaluation of the total irrigable area of the country as a basis for estimating future irrigation potential. The proposed procedures are described in the TOR agreed with NIA in July 1990, and refined in October 1990, and will include both revisions to the irrigable areas estimates summarized in Table 3.1 and the development of systematic adjustment coefficients based on project experience. It is now expected that the results of this exercise, together with detailed supporting maps, will be finalized by NIA in September 1991.

B. PAST IRRIGATION DEVELOPMENT

Historical Development

3.10 Recent publications have described the evolution of irrigation in the Philippines (Svendson et al, 1990; NIA, 1990). The Banawe terraces are witness to its long history. Though classified as communal, Bagadion (1989) estimates that they cover 25,000 ha, representing a complex and impressive engineering achievement. Communal irrigation on a lesser scale was widespread but the first extensive lowland schemes were on lands granted to religious orders by the Spanish king (the 'friarlands'). Some 30 schemes over about 28,000 ha were built close to Manila, on occasion apparently using forced labor. The other important feature of the Spanish period was the cooperative irrigation societies ('zinjara') of the Ilocos region in Northern Luzon. These societies constructed numerous schemes by diverting small rivers that cross the narrow coastal plain. As in the hills they were communal and characterized by complex organizational arrangements to mobilize labor for construction and repair.

3.11 By the end of the Spanish era, perhaps 200,000 ha were irrigated, all in communal or private schemes. With the American administration came the first public involvement. In 1902, the Church sold the friarlands to the new Government; in 1905, a Bureau of Public Works (BPW) was established; and in 1908, a separate Irrigation Division was created. BPW's organization varied but it retained responsibility for irrigation matters until NIA's formation in 1964. The first new public or national system dates to 1913 (the San Miguel scheme in Tarlac) and by 1930 a further eleven systems, covering about 80,000 ha, had been completed. These were mainly in Central Luzon, the largest being on the Angat (27,000 ha) and Pengeranda (17,500 ha) rivers, but some were located in Ilocos and on Panay Island in the Western Visayas.

3.12 Major new development ceased in the 1930s/40s although direct assistance to communal irrigation through the DPW budget dates to this period (no doubt local public support had occurred earlier). With independence came renewed expansion, including support for communal irrigation and the first projects in Mindanao and other new areas. Expansion slowed towards the end of the 1950s but by 1964, when NIA was created, it was estimated that 541,000 ha were being irrigated of which 218,000 ha (40%) came under NIS, 272,000 ha (50%) under CIS and 51,000 ha (9%) under private pump irrigation using surface or groundwater.

The NIA Period

3.13 NIA was established in 1964 with a mandate to create "an irrigation age". Major impetus was provided in the early 1970s by the Bank-supported